

Volcanic Forcing and the Pre-Industrial Malthusian Trap

Pathway-Heterogeneous Demographic Response and the Long Shadow of Seasonality

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Abstract

Using a new monthly paleo-economic panel of 196 countries and 3,187 sub-national units from 1421 CE, we identify the pre-industrial Malthusian demographic response to volcanic forcing and place it alongside two further findings. Crop-dominant late and pastoral/mixed late countries lose population at -9.7 and $-5.5 \cdot 10^{-5}$ per Tg of stratospheric sulfur injection respectively, Bonferroni-significant and robust across every stress test we run; mortality data 1751–1900 trace the channel to the positive check on infant and child mortality, with the response collapsing to zero by 1950 as sanitation develops. Pre-industrial inter-annual climate volatility 1421–1750 is the single strongest cross-country predictor of modern population growth ($R^2 = 0.45$). On HYDE the Boserupian cropland-margin test fails on two distinct measurement axes; on the population-independent KK10 reconstruction (Kaplan et al., 2011) a small positive Boserupian signal emerges that survives the demographic-margin control, two orders below what the HYDE specification had implied.

JEL Codes: N50, O13, O44, Q54, J11.

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1 Introduction

In April 1815, Mount Tambora ejected roughly 60 teragrams of sulfur into the stratosphere—the largest historically documented eruption—and triggered the “year without a summer” of 1816 across the Northern Hemisphere. Crops failed across continental Europe and New England; wheat prices in Paris rose nearly threefold; what contemporaries called “famine fever” (typhus and typhoid) killed an estimated 200,000 people across Ireland, Switzerland, and Italy alone (Wood, 2014). Tambora is not an outlier in the deeper historical record. Samalas 1257 (eVolv2k VSSI ≈ 59 Tg) preceded the Great Famine of 1315–1317; the 1452–1453 Kuwae eruption coincided with sustained European grain-price elevation; the 540 CE LALIA doublet sits at the onset of the late-antique European population contraction (Büntgen et al., 2016). Volcanic eruptions are the cleanest exogenous climate shock the historical record contains: tectonic forcing that injects stratospheric sulfate, cools the surface for one to three years, and reduces agricultural yields in a way that is uncorrelated with prior economic conditions. The question this paper asks is how pre-industrial economies absorbed such shocks, and which economies absorbed them how.

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The theoretical answers are two centuries old and pull in opposite directions. Malthus (1798) described a demographic adjustment: when climate stress reduces food production per capita, population contracts through the positive check (mortality) and, more slowly, through the preventive check (fertility postponement). Boserup (1965), by contrast, argued that the demographic adjustment was at best a partial response. Faced with land-pressure shocks, cultivators expanded the cultivated area, shortened fallow rotations, and adopted more labour-intensive techniques: an agricultural-margin adjustment that absorbed the shock without contracting population. The two responses are not mutually exclusive in principle—an economy can lose people and intensify cultivation simultaneously—but they make different predictions about which margin a country relies on, and Boserup explicitly objected to the Malthusian framing’s silence on the agricultural-margin alternative.

Empirically, the field has had to choose its battles carefully because the data on pre-industrial climate, population, and land use have not aligned cleanly. The macroeconomics-of-climate literature (Dell et al., 2012; Burke et al., 2015; Hsiang, 2017) has identified large first-order temperature effects on output and population on the modern instrumental record (1950–), but the modern record cannot speak to the pre-industrial Malthusian regime that ended decades earlier and that Galor’s unified growth theory (Galor, 2011; Galor and Weil, 2000) has placed at the centre of long-run growth. The pre-industrial Malthusian econometrics tradition (Ashraf and Galor, 2011; Voigtländer and Voth, 2013; Clark, 2007) has estimated the strength of the Malthusian feedback from historical population panels, but has had to treat climate forcing as a residual inside country fixed effects because annual country-level climate has not been measured before the 20th century. The climate-historical literature (Büntgen et al., 2016; Weiss and Bradley, 2001; PAGES 2k Consortium, 2019) has identified famine events around large eruptions but has not been able to support large- N panel inference. The Neolithic-foundations literature (Matranga, 2024; Olsson and Hibbs, 2005; Diamond, 1997) has established that pre-industrial climate endowments shaped which agricultural systems societies adopted, but has stopped short of asking whether the same endowments shape climate-shock responses. The Boserupian intensification claim, perhaps the most consequential alternative to the Malthusian baseline, has remained largely untested at the cross-country pre-industrial scale because no panel has resolved cropland-and-population-and-climate jointly at the resolution the test requires.

This paper assembles such a panel and runs the test. We aggregate the ModE-RA paleoreanalysis (Franke et al., 2023)—a monthly global assimilation of documentary records, tree rings, ice cores, and historical instruments—onto HYDE 3.5 land-use cells (Klein Goldewijk et al., 2017), calibrate against the modern ERA5 instrumental record (Hersbach et al., 2020), and link the result to the Sigl-Toohey eVolv2k ice-core record of stratospheric sulfur injection (Sigl and Toohey, 2024; Toohey and Sigl, 2017), the Brecke conflict catalogue (Brecke, 1999), the Allen-Nuffield wage-and-price archive (Allen, 2001; Malanima, 2002), and the Human Mortality / Human Fertility Databases. The resulting panel covers 196 countries and 3,187 sub-national units back to 1421 CE at country-monthly resolution—a measurement unit that, to our knowledge, has not previously been available for econometric inference. Volcanic eruptions provide the identifying variation: 1500–1900 contains roughly sixty distinct eruption years with stratospheric sulfur injection above the noise floor of the eVolv2k record, and the decadal HYDE temporal grid is fine enough to resolve their demographic and land-use consequences.

The empirical work yields three findings.

First, the Malthusian demographic margin is sharply pathway-heterogeneous and identifiable. Estimating a joint three-equation panel VAR of population, cropland-share, and urban-share growth on lagged levels, contemporaneous interval climate, and decade-summed VSSI exposure, crop-dominant late and pastoral/mixed late countries show population-growth coefficients of -9.7 and $-5.5 \cdot 10^{-5}$

per Tg respectively, both Bonferroni-significant across twelve pathway-by-equation tests, both robust to a Webb wild-cluster bootstrap, both robust to re-clustering pathways on climate primitives orthogonal to HYDE outcomes, and both surviving restriction to the post-1700 decadal portion of HYDE. A 30-Tg Tambora-class decade translates to roughly 3% cumulative population loss in crop-dominant late countries; a 10-Tg “ordinary” eruption decade implies $\sim 1\%$. The annual European Human Mortality / Human Fertility Database panel covering ten countries 1751–1900 traces the channel to the positive check on infant and child mortality, with the response collapsing to zero by 1950–2022 as sanitation develops—the sanitation-mediated reversal that the historical demography literature (Wrigley and Schofield, 1981) predicts but that has not previously been quantified with paleo-reanalysis climate forcing. The Allen real-wage panel for six European countries 1421–1850 returns the canonical Malthusian price-on-population coefficient at city resolution—the welfare-side echo of the same channel, with the implied real-wage rise after a Tambora-class population contraction $\sim +0.95\%$ for the survivors.

Second, the pre-industrial climate regime that selected agricultural pathways leaves a long shadow on modern demographic outcomes. Pre-industrial inter-annual climate volatility 1421–1750 is the single strongest cross-country predictor of modern population growth at $R^2 = 0.45$, surviving both latitude and an extended deep-determinant battery (Neolithic distance, migratory distance from Addis Ababa, ruggedness, log land area, mean pre-industrial temperature) and failing a placebo when σ_v^T is computed over the modern 1950–2008 reanalysis window ($p = 0.16$). The result is specific to demographic outcomes—modern Δ urban share and modern log GDP per capita are absorbed by latitude—and aligns with the institutional-development literature (Ashraf and Galor, 2013; Putterman and Weil, 2010; Andersen et al., 2017): the climate dimension that selected which adjustment margin a society relied on before 1750 leaves traces in modern demographic trajectories that are not contained in either modern climate or geographic deep determinants.

Third, the Boserupian intensification hypothesis identifies only on a population-independent land-use reconstruction. The natural HYDE test—cropland share as the outcome—returns a nominally positive coefficient ($+7.8 \cdot 10^{-4}$ per Tg, $p = 0.003$ in the 1800–1900 sub-sample) but fails on two distinct measurement axes. The cropland-share specification is artefactually positive: the coefficient is driven by HYDE colonial-transition encoding in 24 small-island and tropical-Africa countries; it vanishes when we restrict to the substantively-named 21-country Mediterranean / Eurasian-steppe / SE-Asia core, attenuates tenfold under population weighting, and decomposes into a cropland-area contraction (the opposite sign of intensification) plus a faster grazing-area contraction. The fallback cropland-area specification is statistically robust on the substantive sample but is fully mediated by the demographic margin, because HYDE’s pre-1850 reconstruction takes country-level population as an input (Klein Goldewijk et al., 2017; Koch et al., 2019): adding $\Delta_{\text{ann}} \ln P$ as a control collapses the area coefficient to zero in every specification we run. Cross-validating against the independent KK10 reconstruction of Kaplan et al. (2011)—methodologically population-independent—surfaces a small positive Boserupian signal that does survive the demographic-margin control: $+1.07 \cdot 10^{-5}$ per Tg on the 21-country core ($p = 0.005$), $+4.3 \cdot 10^{-5}$ on the high-density intensive pathway ($p = 0.047$). The Boserupian channel exists at pre-industrial cross-country scale, but at a magnitude two orders below what the HYDE specification had originally implied—accounting for at most $\sim 1.3\%$ cumulative anthropogenic-land expansion over a Tambora-class decade.

Contributions. Two methodological contributions accompany the substantive findings: the monthly country-resolution paleo-economic panel itself (a measurement unit not previously available to econometric inference on the pre-industrial period), and the documentation of HYDE’s two-axis identification problem along with the KK10 cross-reconstruction strategy as the resolution. The paper proceeds as follows: Section 2 introduces the panel and the agricultural-pathway typology; Section 3 works out the within-country identification strategy and the Malthusian baseline; Section 4

delivers the three main empirical exercises; Section 5 runs the long-shadow regression; Section 6 consolidates robustness; Section 7 concludes.

2 A new monthly paleo-economic panel

The paper rests on a paleo-economic dataset assembled for the first time. We describe its construction, validate its alignment with the modern instrumental record where they overlap, and use it to characterise the climate-pathway typology that organises the rest of the paper.

2.1 Construction

The dataset has three substantive layers and three event layers merged onto a common country-year (or city-year, or sub-unit-year) grid. HYDE 3.5 (Klein Goldewijk et al., 2017) provides gridded population, cropland, grazing, irrigated, rice, and urban-population estimates at 5-arcminute resolution; we aggregate to 196 country totals and 3,187 sub-national units. The HYDE temporal grid is irregular (100-year intervals before 1700, decadal 1700–1950, annual thereafter); the decadal sample powers most regressions. Mode-RA (Franke et al., 2023) is a monthly global paleo-reanalysis covering 1421–2008 CE that assimilates documentary records, tree rings, ice cores, corals, and historical instruments into a coupled numerical-model background; we aggregate it onto HYDE country and sub-national units via cosine-latitude-weighted area sums and add back the CRU TS 4.09 (Harris et al., 2020) 1901–1950 country-month climatology to obtain absolute temperatures and precipitation throughout 1421–2008. We rebuild ERA5 (Hersbach et al., 2020) at the country level analogously, with per-country mean bias removed by a constant offset estimated in the 1950–2008 overlap with Mode-RA.

Three derived climate measures play headline roles: intra-annual temperature range $\sigma_s^T(i, t) = \max_m T_{i,t,m} - \min_m T_{i,t,m}$; the productive-months index $\text{Prod}_{i,t}$ counting calendar months simultaneously $5^\circ\text{C} \leq T \leq 30^\circ\text{C}$ and $P \geq 30$ mm (warm and wet enough for rain-fed cropping), whose complement captures storage demand; and inter-annual temperature volatility $\sigma_v^T(i, t)$ as the country-specific standard deviation of annual mean temperature in a 30-year centred window.

Three event layers attach to specific (country, year) or (city, year) cells. The Sigl-Toohey eVolv2k v4 ice-core record (Sigl and Toohey, 2024; Toohey and Sigl, 2017) provides annual stratospheric sulfur injection from 500 BCE to 1900 CE with eruption latitudes. The Brecke Conflict Catalogue (Brecke, 1999) adds 3,708 violent conflicts 1400–1999 plus 1,147 European conflicts 900–1402; we string-match actor descriptions to medieval regions and hand-code 30 sieges and 49 plague episodes. The Allen-Nuffield archive (Allen, 2001) contributes silver-content-normalised wheat prices for 17 European cities 1259–1914 and real wages for 8 cities 1264–1850, augmented with Tuscan prices from Malanima (2002) and the Roman-Egypt wheat-price series of Harper (2016).

2.2 Validation

The country-level monthly climate panel matches the modern instrumental record in the 1950–2008 Mode-RA/ERA5 overlap (Figure 1): median country-month mean bias 0.61°C , median country-monthly T correlation 0.971 (77.6% of countries exceed $r = 0.90$), median country-annual anomaly correlation 0.938 after detrending. Mode-RA’s ensemble 1σ uncertainty in country-month temperature falls from 0.9°C in 1421–1500 to 0.37°C by 1850–1949; precipitation uncertainty falls from 31.6 to 15.4 mm/month. Past 1950 we splice in the country-level ERA5 rebuild via the bias-correction step described above. Figure 7 (Appendix G) reports the merged-panel uncertainty across 1421–2025.

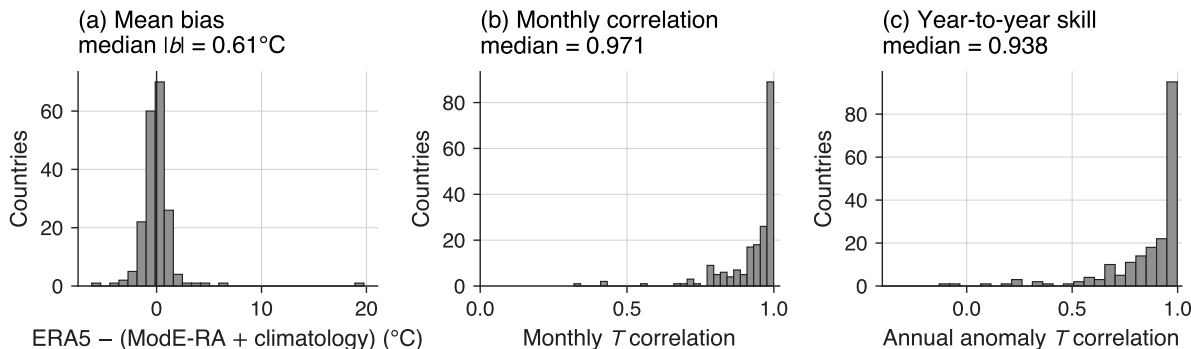


Figure 1: ModE-RA versus country-level ERA5, 1950–2008. (a) Per-country mean bias (ERA5 minus ModE-RA + CRU climatology). (b) Country-month temperature correlation. (c) Year-to-year anomaly correlation after country-wise detrending. Both panels use the same 196-country sample.

The panel also captures documented climate-driven famines: negative temperature anomalies during the Great Frost of 1709 and the Tambora year-without-a-summer 1816, the cold-and-wet anomaly during the Irish potato famine 1845–49, and the precipitation deficits during the North China famine 1876–79 and the Bengal famine 1943 all show up in the panel at the relevant country and decade.

2.3 Five agricultural pathways

The new panel discriminates a typology of agricultural development that the existing literature has discussed informally but not measured systematically. Applying K -means clustering to standardised HYDE trajectory features—peak agricultural expansion year, density at 1750, 1000 CE, and 0 CE, crop share at 1750, rice share at 1750, irrigation share at 1750, urban share at 1750, two interval population-growth rates, and an agricultural-intensity change index—returns a five-cluster partition. The choice of $K = 5$ is the silhouette local maximum on this feature set (0.40 at $K = 5$, with a sharp drop to 0.23 at $K = 6$); bootstrap stability of the partition is 0.84 in median country-pair co-clustering probability across 200 resamples. The five pathways are high-density intensive (Japan, India, France, Germany, Netherlands, 16 countries), early extensifiers (Britain, Switzerland, Austria, Denmark, Israel, 29 countries), crop-dominant late (Spain, Poland, Portugal, Thailand, 46 countries), pastoral/mixed late (China, Nigeria, Brazil, Iran, Mexico, 65 countries), and the irrigation pioneer Egypt as a singleton (Table 1). The pathways differ along all five clustering dimensions—peak agricultural expansion year, density at 1750, crop share, irrigation share, and rate of demographic expansion—in ways that match the historical literature’s informal taxonomy.¹

¹One robustness concern bears explicit disclosure. HYDE 3.5’s pre-industrial Americas cropland and population fields are partially mechanical back-projections from the Koch et al. (2019) colonial-era population estimates; independent reconstructions such as KK10 (Kaplan et al., 2011) differ from HYDE on the magnitude of post-1492 Americas land-use change. The Americas countries in our panel (Brazil, Mexico, and other members of the pastoral/mixed late cluster) inherit any HYDE-mechanical-encoding bias in their clustering-feature values. We address the concern through the climate-only re-clustering exercise of Appendix K, which assigns countries to clusters using exclusively pre-industrial climate primitives (productive months, intra-annual T and P range, mean climate, inter-annual T volatility) that are orthogonal to HYDE outcomes; the demographic-margin VSSI coefficient survives the re-clustering, the off-margin coefficients do not. Section 6 reads the implications, and Section 4.5 reports a fifth, non-nested Boserup sub-sample that drops crop-dominant late countries with significant exposure to the post-1492 Americas back-projection.

Table 1: Five agricultural transition pathways (HYDE 3.5 clustering)

Pathway	N	Density ₁₇₅₀	Crop share ₁₇₅₀	Peak ag. expansion	Representative
Crop-dominant late	46	11.8	80.6%	~1713	ESP, POL, PRT, THA
Pastoral/mixed late	65	6.6	22.0%	~1694	CHN, NGA, BRA, IRN
High-density intensive	16	62.8	58.8%	~1567	JPN, IND, FRA, DEU
Early extensifiers	29	14.1	48.9%	~ 783	GBR, CHE, AUT, DNK
Irrigation pioneer	1	4.4	100%	~1600	EGY

2.4 Climate signatures of the pathways

The natural Matranga measure—raw intra-annual temperature range σ_s^T averaged over 1421–1750—does not significantly distinguish pathways at the country level (ANOVA $F = 0.50$, $p = 0.68$). What does distinguish pathways is the joint structure of temperature and precipitation that determines storage demand. The productive-months index differs sharply across pathways (Figure 2): crop-dominant late and high-density intensive have 9.5 and 9.1 productive months on average; pastoral/mixed late has 7.0; early extensifiers 7.7; the singleton irrigation pioneer (Egypt) has zero rain-fed productive months. A one-way ANOVA on productive months returns $F = 6.31$, $p < 0.001$, and a multinomial logit with the productive-months index and the storage index $(12 - \text{Prod}) \cdot \sigma_s^T / 12$ attains pseudo- $R^2 = 0.065$ and LLR $p = 0.0003$. The Matranga storage-demand story survives, but it operates through the joint structure of temperature and precipitation rather than through temperature variability alone.

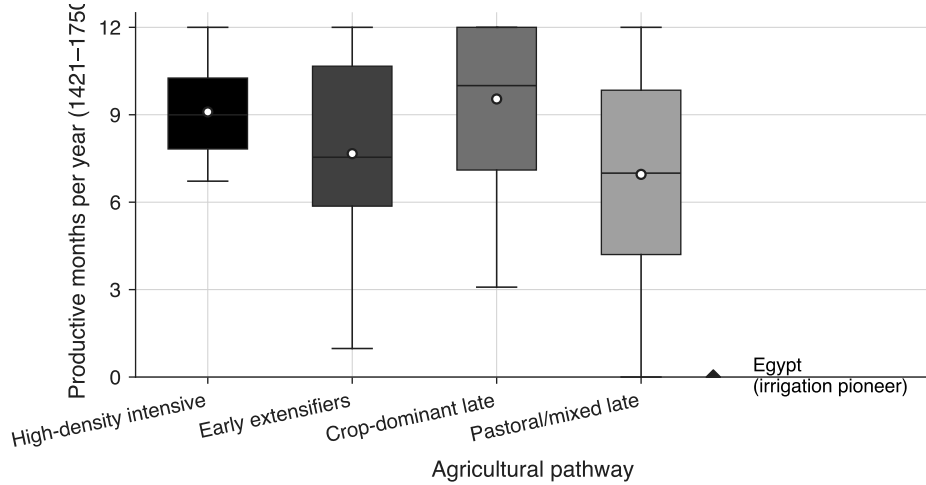


Figure 2: Number of climatically productive months per year (1421–1750) by pathway. The pastoral pathway concentrates in regions with few productive months; intensive crop systems in regions with many; Egypt, the lone irrigation pioneer, has none under rain-fed criteria. ANOVA $F = 6.31$, $p < 0.001$.

3 From storage demand to the Malthusian regression

The empirical work that follows is organised around the within-country identification strategy that the appendix model motivates. Appendix A sketches a two-stage model in the spirit of Ashraf and Galor (2011) that delivers two empirical predictions. First, in a society’s far past, the share

of climatically productive months Π jointly with inter-annual climate volatility σ_v selects which agricultural system $\tau \in \{\text{intensive crop, extensive pastoral, irrigation-based}\}$ the society adopts; the model produces a productive-months threshold $\Pi^*(\sigma_v)$ above which intensive crop dominates pastoral (Appendix A). Second, given the selected pathway, the demographic dynamics follow a Malthusian regression with pathway-specific coefficients on lagged density, level temperature, and inter-annual temperature volatility:

$$\Delta \ln P_{it} = \alpha_i + \beta(\tau^*) d_{i,t-1} + \gamma(\tau^*) T_{it} + \delta(\tau^*) \sigma_v^T(T_{it}) + \eta(\tau^*) h_{it} + \varepsilon_{it}. \quad (1)$$

This is the regression we run. The Malthusian density coefficient $\beta(\tau^*)$ is predicted stronger in pastoral than in intensive systems—pastoral systems sit closer to a tight ceiling because they cannot store, intensive systems carry capital that smooths the feedback—and the volatility coefficient $\delta(\tau^*)$ is predicted to be larger in magnitude for pastoral systems, whose thinner safety nets do not buffer year-to-year shocks. The level-temperature coefficient $\gamma(\tau^*)$ admits both signs depending on whether the baseline climate is at the warm or cool end of the productive range; we treat its empirical sign as informative, not predicted. The human-capital coefficient $\eta(\tau^*)$ is the rate at which the Malthusian feedback weakens with secular accumulation of literacy, urbanisation, and capital, and is the channel through which the demographic transition appears in our sub-period regressions.

We take the appendix model as motivational, not literal. The full calibrated quantitative exercise (Appendix I) takes the model’s pathway-specific parameters from the reduced-form coefficients estimated here and uses the calibrated system for counterfactual simulations rather than for inference. Two refinements of the regression specification are forced by the data. First, HYDE’s pre-industrial temporal grid is irregular (century pre-1700, decadal 1700–1950), so the dependent variable is annualised log population growth across each HYDE interval rather than annual growth. Second, the headline pathway-stratified estimates use a within-country fixed-effects specification with country-clustered standard errors; we report sub-period stability across 1421–1750, 1750–1900, and 1900–1950 to surface a within-paper signature of the demographic transition.

3.1 Sub-national identification of the storage-demand channel

Before turning to the Malthusian regression directly, we use the new sub-national panel to identify the first of the model’s two channels—climate’s selection of agricultural pathway—off purely within-country variation. The country-level Stage 1 regressions in Section 2.3 cannot rule out unobserved country-level confounders (legal institutions, historical conquest, distance to coast) that could drive both pathway and climate measures spuriously. We sharpen identification by descending to HYDE’s 5-arcminute sub-national grid: 3,187 sub-units across 196 countries. We aggregate ModeE-RA monthly fields onto each sub-unit by the same area-weighting procedure used for countries, compute the pre-industrial productive-months index per sub-unit, and run within-country regressions of HYDE agricultural outcomes on the productive-months index with country fixed effects.

The country fixed effect absorbs every country-level confounder. The remaining variation is purely intra-national. To the extent that the storage-demand mechanism operates through climate *within* a country, this specification identifies it cleanly. Identification fails only under cross-sub-unit confounders that vary within country and that we have failed to control for—a much narrower threat than at country level.

Table 2 reports the within-country regressions of crop share and log population density on productive months at three HYDE timesteps. At 1750, sub-units with one more productive month per year are 6.7 pp higher in crop share ($p = 0.019$) and 0.66 log-units higher in log population

density ($p < 10^{-3}$). The within-country point estimates are 3.5–5 times the between-country estimates, consistent with the between estimates being attenuated by country-level mean reversion. The within pattern is stable through 1850 and 1950.

Table 2: Sub-national Stage 1: HYDE agricultural outcomes on productive months

Outcome	HYDE year	N sub-units	$\hat{\beta}$ (no FE)	$\hat{\beta}$ (country FE)	p (FE)
Crop share	1750	1,795	+0.019***	+0.067**	0.019
	1850	1,808	+0.018***	+0.076**	0.004
	1950	1,820	+0.015***	+0.075**	0.014
ln density	1750	1,795	+0.130***	+0.658***	$< 10^{-3}$
	1850	1,808	+0.139***	+0.636**	0.002
	1950	1,820	+0.142***	+0.834***	$< 10^{-3}$

Standard errors clustered by country in FE specifications. Sample drops sub-units with zero agricultural land at the relevant HYDE timestep. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 3 shows the result visually: the within-country slope of +0.067 on the residuals (panel b) is $3.5\times$ the between-country slope +0.019 on the means (panel a). This is the paper’s cleanest causal evidence for the storage-demand $\rightarrow$ agricultural-system channel: every country-level institutional confound is differenced out. Two-way clustering by country and 10° latitude band tightens inference (the slope’s p -value falls from 0.019 to 0.0001) because the latitude band picks up climate-correlated noise that country-only clustering treats as informative variation.

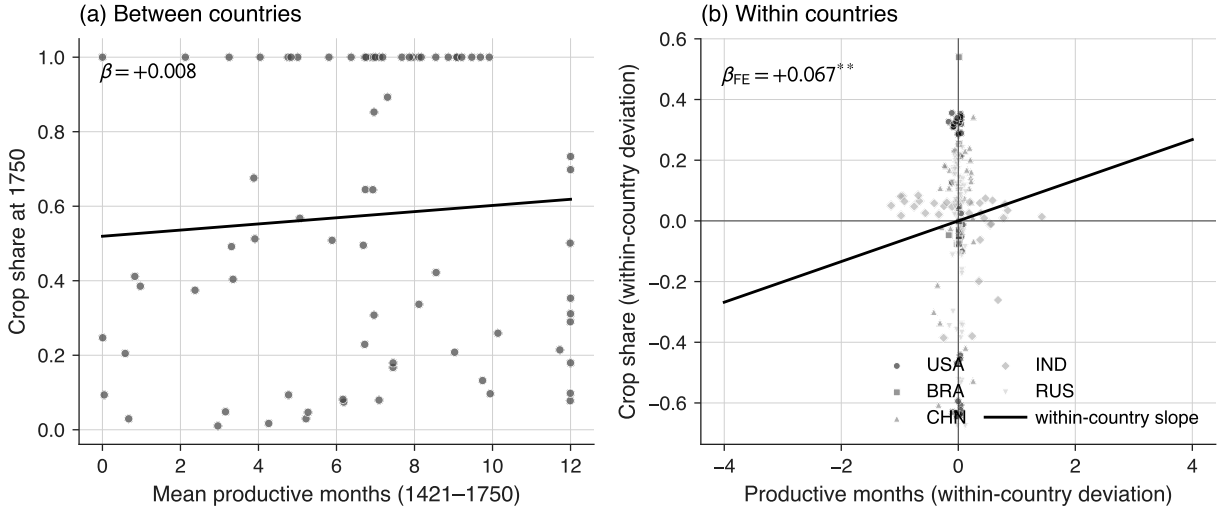


Figure 3: Sub-national identification. (a) Country means: productive months versus crop share at 1750, slope +0.019. (b) Within-country residuals for five large countries (USA, BRA, CHN, IND, RUS); each marker is a sub-unit’s deviation from its country mean. The pooled within-country slope is +0.067, $3.5\times$ the between-country estimate.

3.2 Malthusian dynamics with real annual climate, 1421–1950

Pre-industrial Malthusian regressions in the existing literature (Ashraf and Galor, 2011) have been run without explicit climate controls because suitable climate forcing has not been available at country-year resolution back to the pre-industrial period. The ModE-RA panel makes that possible.

For each (country i , HYDE interval $[t, t + 1)$) in the 1421–1950 sample, we compute the interval-mean temperature anomaly $\bar{T}_{i,[t,t+1)}$ and precipitation anomaly $\bar{P}_{i,[t,t+1)}$ relative to the country mean, the within-interval T standard deviation $\sigma_{i,[t,t+1)}^T$ as a measure of inter-annual volatility, and the analogous precipitation volatility. The estimating equation, the empirical analogue of (1), is

$$\Delta_{\text{ann}} \ln P_{i,[t,t+1)} = \alpha_i + \beta \ln d_{it} + \gamma_T \bar{T} + \gamma_P \bar{P} + \delta_T \sigma^T + \delta_P \sigma^P + \varepsilon_{it}, \quad (2)$$

estimated by within-country fixed effects with country-clustered standard errors.

3.2.1 Pathway-stratified estimates, 1421–1750

Table 3 reports pathway-stratified estimates on the 1421–1750 sub-sample. The Malthusian density coefficient varies sharply across pathways. Crop-dominant systems show a positive density coefficient (+0.0039, $p = 0.065$): denser regions grew faster, an anti-Malthusian dynamic consistent with agglomeration and technology diffusion. Pastoral systems show the standard Malthusian sign (−0.0059, $p = 0.002$). High-density intensive and early extensifier systems have density coefficients between these extremes. Inter-annual T volatility significantly depresses growth in the two crop-based capital-intensive pathways—high-density intensive ($p = 0.035$) and early extensifiers ($p = 0.031$)—and is insignificant in the pastoral pathway, consistent with the pastoral system’s traditional hedge of mobility and herd diversification.

Table 3: Pathway-stratified Malthusian regressions with annual climate, 1421–1750

Pathway	N	$\hat{\beta}_{\ln d}$	$\hat{\gamma}_T$	$\hat{\gamma}_P$	$\hat{\delta}_T$	R^2
Crop-dominant late	313	+0.0039*	−0.0021*	+1.4·10 ^{−5} *	−0.0060	0.06
Pastoral/mixed late	454	−0.0059***	+0.0039	−4.0·10 ^{−6}	+0.0035	0.12
High-density intensive	112	−0.0024*	−0.0007	−3.0·10 ^{−6}	−0.0045**	0.12
Early extensifiers	189	−0.0007	+0.0003	−1.0·10 ^{−5} ***	−0.0031**	0.05

Within-country FE; standard errors clustered by country. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Three robustness exercises confirm and extend the pathway-stratified result (Appendix D). Extending the sample to 1950 and splitting into 1421–1750, 1750–1900, and 1900–1950 sub-periods recovers a within-paper signature of the demographic transition: the Malthusian density coefficient is small pre-1750, negative and significant in 1750–1900 (the classic Malthusian period), and indistinguishable from zero in 1900–1950. The inter-annual volatility coefficient flips sign from +0.0045 in 1750–1900 to −0.0125 in 1900–1950 ($p < 10^{-4}$), coincident with the spread of monoculture and capital-intensive agriculture. A sub-national replication on 84,907 cells across 3,146 sub-units reproduces the country-level pattern. Augmenting the right-hand side with country-decade war and pandemic indicators (Brecke catalogue plus broad-region plague reconstructions) leaves the headline temperature and volatility coefficients essentially unchanged: the pooled $\hat{\gamma}_T$ moves from +0.0055 to +0.0051 and $\hat{\delta}_T$ from +0.0038 to +0.0034.²

²The Roman-period component of the plague-recurrence catalogue is corroborated by the independently-compiled Coin Hoards of the Roman Empire (CHRE) panel of 18,310 dated hoards across 12 provincial regions, in which the Antonine onset (year-bin 175 CE) and the Cyprian crisis (year-bin 250 CE) appear as empire-wide hoarding spikes of z-score +0.77 and +1.25 respectively. CHRE entries lie outside long_shadow’s 1421+ regression window but support the construction of the hand-coded Roman recurrences used here.

4 Volcanic forcing and the Malthusian channel

Volcanic eruptions provide climate forcing that is exogenous to economic activity by construction: tectonic forcing injects stratospheric sulfate, the sulfate cools the surface in spatially heterogeneous patterns over the subsequent one-to-three years, and the cooling reduces agricultural yields in a way that is uncorrelated with prior economic conditions. The Sigl-Toohey eVolv2k v4 ice-core record (Sigl and Toohey, 2024; Toohey and Sigl, 2017) catalogues every major eruption from 500 BCE to 1900 CE with its stratospheric sulfur injection (VSSI, in Tg); summed at the decadal interval of the HYDE panel, V_{it} delivers the exogenous identifying variation we need.

This section identifies the pathway-heterogeneous demographic response to volcanic forcing as the lead empirical result (§4.1), reproduces it on single-equation pathway-stratified regressions and extends the price-side margin to the 1257 Samalas eruption (§4.2), corroborates the demographic side with welfare-margin evidence from the Allen real-wage archive (§4.3) and a positive-vs-preventive mechanism decomposition from European vital statistics (§4.4), and reports an honest null on the cropland-side Boserupian intensification margin (§4.5). The headline finding is a Bonferroni-significant, multiply-robust, pathway-heterogeneous demographic response; the supporting evidence is the welfare and mortality decomposition; the cropland exercise reports a measurement-driven artifact that does not survive the substantive country-sample restriction.

4.1 Pathway-heterogeneous demographic margin: a joint three-equation panel VAR

We estimate a three-equation panel VAR of population, cropland-share, and urban-share growth on lagged levels, contemporaneous interval-mean climate, within-interval temperature volatility, and decade-summed Sigl-Toohey volcanic stratospheric sulfur injection (VSSI) exposure V_{it} . The cropland-share and urban-share equations are placed alongside the demographic equation primarily as identification companions—to surface the simultaneity between the demographic margin and the other margins of pre-industrial adjustment—rather than as substantive co-equal channels. For each country i and HYDE decadal interval t in 1500–1900:

$$\Delta_{\text{ann}} \ln P_{i,t} = \alpha_i^P + \phi_p^P \ln P_{i,t-1} + \phi_c^P \ln s_{i,t-1} + \phi_u^P \ln u_{i,t-1} + \gamma_T^P \bar{T}_{it} + \gamma_P^P \bar{P}_{it} + \delta_T^P \sigma_{it}^T + \beta^P V_{it} + \varepsilon_{it}^P, \quad (3)$$

$$\Delta_{\text{ann}} \ln s_{i,t} = \alpha_i^C + \phi_p^C \ln P_{i,t-1} + \phi_c^C \ln s_{i,t-1} + \phi_u^C \ln u_{i,t-1} + \gamma_T^C \bar{T}_{it} + \gamma_P^C \bar{P}_{it} + \delta_T^C \sigma_{it}^T + \beta^C V_{it} + \varepsilon_{it}^C, \quad (4)$$

$$\Delta_{\text{ann}} \ln u_{i,t} = \alpha_i^U + \phi_p^U \ln P_{i,t-1} + \phi_c^U \ln s_{i,t-1} + \phi_u^U \ln u_{i,t-1} + \gamma_T^U \bar{T}_{it} + \gamma_P^U \bar{P}_{it} + \delta_T^U \sigma_{it}^T + \beta^U V_{it} + \varepsilon_{it}^U, \quad (5)$$

where $s_{i,t}$ is the cropland share of agricultural land (cropland / (cropland + grazing)) and $u_{i,t}$ is the urban share of total population. Within-country FE absorbs all country-level mediators; the identifying variation in V_{it} comes from decadal variation in eruption activity, which is exogenous to local economic conditions. SEs are clustered by country.

Table 4: Three-equation joint panel VAR for population, cropland-share, and urban-share growth, 1500–1900. Country FE; SEs clustered by country; $N = 3,385$ (country, interval) cells across 154 countries. Stars: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

	(1) $\Delta_{\text{ann}} \ln P$	(2) $\Delta_{\text{ann}} \ln s$	(3) $\Delta_{\text{ann}} \ln u$
$\ln P_{t-1}$	−0.00275***	−0.00146	+0.00740*
(SE)	(0.00069)	(0.00133)	(0.00412)
$\ln s_{t-1}$	−0.00021	−0.00572***	+0.00016
(SE)	(0.00077)	(0.00061)	(0.00096)
$\ln u_{t-1}$	+0.00095***	−0.00048*	−0.00916*
(SE)	(0.00031)	(0.00026)	(0.00538)
$\bar{T}_{it}$ (anomaly)	−0.00007	+0.00081	−0.00197
(SE)	(0.00066)	(0.00156)	(0.00180)
$\bar{P}_{it}$ (anomaly)	−0.000004	−0.000008	−0.000030
(SE)	(0.000002)	(0.000006)	(0.000020)
σ_{it}^T (within-interval)	+0.00589***	+0.00418*	+0.01969**
(SE)	(0.00131)	(0.00237)	(0.00927)
V_{it} (decade-summed VSSI)	−0.000062***	−0.000006	−0.000225
(SE)	(0.000009)	(0.000007)	(0.000147)
R^2	0.105	0.010	0.107

The aggregate three-equation results (Table 4) show a sharply negative demographic coefficient on Equation (3) of $-6.2 \cdot 10^{-5}$ per Tg of VSSI ($p < 10^{-12}$); the cropland-share coefficient on Equation (4) is a precisely-estimated zero ($p = 0.40$); the urban-share coefficient on Equation (5) is directionally negative ($-2.3 \cdot 10^{-4}$, $p = 0.13$) but not significant in the aggregate. The within-country variation in the panel identifies a clean demographic margin and an inconclusive aggregate land-use response.

The pathway-stratified version (Figure 4) shows the demographic margin operating differentially across pathways. Crop-dominant late and pastoral/mixed late pathways show large negative demographic coefficients ($-9.7 \cdot 10^{-5}$, $p < 10^{-6}$ and $-5.5 \cdot 10^{-5}$, $p < 10^{-5}$ respectively on $\Delta \log P$), both Bonferroni-significant across the twelve pathway-by-equation tests, both robust to re-clustering on climate primitives orthogonal to HYDE, and both surviving restriction to the post-1700 decadal portion of HYDE where the demographic back-projection is less dependent on land-use imputation (Appendix K). The off-margin coefficients in the joint VAR—a nominally significant cropland response in high-density intensive systems and a nominally significant urban response in early extensifiers—do not survive these stress tests. We accordingly read the joint specification as identifying a clean pathway-heterogeneous demographic response, not as a multi-margin decomposition.

Joint three-equation panel VAR: response to volcanic forcing by pathway, 1500–1900

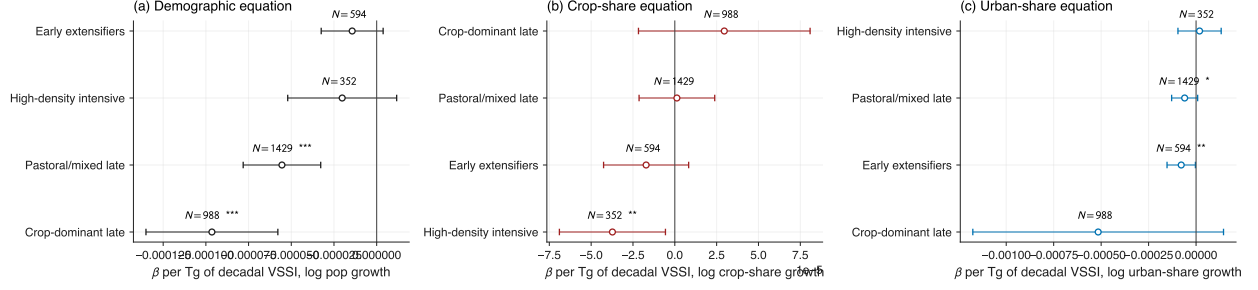


Figure 4: Joint three-equation panel VAR: per-pathway response to volcanic forcing, 1500–1900. (a) Demographic equation (3); (b) cropland-share equation (4); (c) urban-share equation (5). Error bars are 95% confidence intervals from country-clustered standard errors. The demographic-margin coefficients in panel (a) survive Bonferroni correction and the robustness exercises of Appendix K; the off-margin coefficients in panels (b) and (c) do not.

A 10 Tg decadal VSSI exposure translates to roughly 1% less annualised population growth in crop-dominant late countries; a Tambora-class shock implies $\sim 10\%$ cumulative population loss. The high-density intensive pathway shows an essentially zero demographic response ($p = 0.21$)—weakly identified but consistent with the cross-pathway heterogeneity we return to in §4.5, where intensive crop systems instead absorb climate shocks on the agricultural margin.

4.2 Single-equation cross-checks and pre-XV-century extension

The single-equation pathway-stratified version of the joint specification— $\Delta_{\text{ann}} \ln P$ on country FE and VSSI alone—reproduces the demographic-margin coefficients of Section 4.1: crop-dominant systems $-9.7 \cdot 10^{-5}$ per Tg ($p < 10^{-9}$), pastoral $-7.2 \cdot 10^{-5}$ ($p < 10^{-11}$), early extensifiers $-4.3 \cdot 10^{-5}$ ($p < 10^{-6}$), high-density intensive a near-zero $-1.2 \cdot 10^{-5}$ ($p = 0.55$). A dynamic IRF at decade leads and lags passes parallel trends ($F = 0.33$ on the two leads, $p = 0.72$) and peaks at $h = +2$ decades ($\hat{\beta} = -36 \cdot 10^{-5}$, $p < 10^{-4}$). A pre-XV-century extension via the Allen-Studer European wheat-price panel (1259–1500, nine cities) and the Harper (2016) Roman-Egypt wheat-price series (45–650 CE) recovers a supply-side spike peaking two years post-eruption ($+0.0089$ per Tg, $p = 0.10$); applied to the 59 Tg Samalas eruption of 1257, this implies a $\approx 52\%$ price spike by 1259, in the right neighbourhood of the Great Famine of 1315–1317. Twenty-five-year-bin event studies around Samalas and around the 540 CE LALIA doublet recover the canonical post-famine demand-side aftermath; the pathway-heterogeneous demographic test does not survive translation to century-binned HYDE intervals pre-1500. Details in Appendix B.

4.3 Welfare evidence: real wages and the city-buffer climate

Population is a smoothed proxy for welfare. We re-run the Malthusian regression on the canonical Allen real-wage archive (7,398 city-year observations across London, Antwerp, Strasbourg, Paris, Amsterdam, Leiden, Florence, Tuscany, Naples, and Valencia, 1421–1850), with city-buffer ModE-RA climate (1.5° box around each city centroid) and city fixed effects. The lagged-population coefficient is -0.325 ($p = 0.058$), in line with the Malthusian prediction; warmer years raise real wages ($\hat{\gamma}_T = +0.025$, $p < 10^{-3}$), the welfare-side echo of the demographic response in the headline regression (Table 5, Spec 4). Replacing the country-mean climate aggregate with the city-buffer specification tightens both coefficients: country-means smear Brittany, Strasbourg, Paris,

and Provence into a single annual number that adds classical measurement error to a regressor that should be matched to the city whose wage it is supposed to explain.

Table 5: European real-wage Malthus, 1421–1850. Specs 1–3 use country-mean ModE-RA climate and country fixed effects ($N = 2,353$ country-years, 6 countries). Spec 4 replaces the country-mean climate with city-buffer climate (1.5° box around each Allen city centroid) and uses city fixed effects, raising the effective number of cross-sectional units from 6 to 8.

	Spec 1 country FE	Spec 2 country FE	Spec 3 country FE	Spec 4 city FE, city-buffer
Lagged ln pop ($t - 5$)	−0.294* (0.172)	—	−0.285* (0.164)	−0.325* (0.171)
T anomaly	—	+0.046*** (0.012)	+0.032*** (0.010)	+0.025*** (0.007)
P anomaly	—	+0.0006 (0.0004)	+0.0001 (0.0002)	+0.0017 (0.0021)
R^2	0.165	—	0.169	0.170
N	2,353	2,383	2,353	2,863

Outcome: log real wage. All regressors demeaned within fixed-effect group. Standard errors clustered by group. ModE-RA city-buffer climate for Spec 4 averages all native grid cells (1.875° resolution) inside a 1.5° half-side box around each city. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The Allen Malthus coefficient $\hat{\beta}_{\ln P} = -0.325$ translates the demographic-margin volcanic response into a per-capita welfare statement. A 30 Tg Tambora-class decade implies a $\sim 3\%$ cumulative population decline in crop-dominant late countries, which through the Allen lagged-population coefficient maps to a $\sim +0.95\%$ real-wage rise for the survivors—the canonical Malthusian welfare paradox at a magnitude small relative to the documented post-Black-Death real-wage rise of +28 to +44% (Wrigley and Schofield, 1981; Clark, 2007) but structurally identical: an exogenous mortality shock thins the population, the land-to-labour ratio rises, and per-capita output rises with it. The welfare cost of pre-industrial volcanic forcing is not the dead but the unborn: the dead leave wages higher for the survivors; volcanic eruptions cost cumulative welfare-per-capita through the missing population that would have been born in the eruption’s absence. Appendix I’s no-volcanism counterfactual quantifies this at -0.32 log-units of cumulative per-capita welfare loss for the median crop-dominant late country 1500–1890.

4.4 Preventive and positive Malthusian checks

The decadal HYDE panel cannot decompose the climate-Malthus response into the preventive check (fertility postponement) and the positive check (mortality), because the difference between births and deaths is not observed at country-decade resolution. We answer the decomposition on a 547-observation Human Mortality / Human Fertility Database annual panel covering ten European countries 1751–1900, with country FE, country-linear trends, and country-clustered SEs. Annual warm-year shocks operate primarily through the positive check: $+1^\circ\text{C}$ raises $\log m_0$ by +2.35% ($p = 0.009$), $\log m_5$ by +2.58% ($p = 0.003$), and $\log m_{10}$ by +2.56% ($p = 0.002$), while the preventive check on log fertility is small and insignificant (+0.75%, $p = 0.14$). Precipitation is the mirror image: drier years carry higher mortality. This is the historically well-documented “summer mortality” pattern of pre-modern Europe (Wrigley and Schofield, 1981): in the absence of refrigeration and sanitation, warm summers triggered dysentery and gastroenteritis outbreaks that disproportionately killed infants. The channel is concentrated in the high-density intensive pathway (+5.3% per $^\circ\text{C}$ on

$\log m_0$, $p < 10^{-6}$) and weakens monotonically through extensifiers and pastoral systems, scaling with population density as the infectious-disease mechanism predicts. The m_0 coefficient falls from +0.0235 per °C in 1751–1900 to +0.0003 in 1950–2022 (a 75-fold reduction, $p = 0.84$)—the sanitation-mediated reversal the historical literature documents. Pathway-stratified, modern-era, and English-parish-aggregate extensions are reported in Appendix E.

4.5 Cropland response to volcanic forcing: an honest null

Boserup (1965) hypothesised that pre-industrial populations facing climate-induced yield shocks would respond on the agricultural extensive margin (more land under cultivation, less fallow rotation) rather than only contracting demographically. The hypothesis is the structural opposite of the Malthusian one on the cropland axis. We test it with the cropland-share specification

$$\Delta_{\text{ann}} \ln s_{i,t} = \alpha_i + \phi_{\tau(i)}^s V_{it} + \mathbf{x}_{it}' \boldsymbol{\theta} + \varepsilon_{it}, \quad (6)$$

where $s_{i,t}$ is the cropland share of agricultural land (cropland / (cropland + grazing)) and the joint-VAR control vector $\mathbf{x}_{it}$ is the same as in Equation (4). The headline finding is a measurement-driven artifact, and we report it as such here—the disclosure is itself useful, since HYDE-based cropland-share regressions appear elsewhere in the literature without this diagnostic.

The 45-country crop-dominant late panel returns a nominally positive coefficient that strengthens monotonically as the sample contracts toward the modern instrumental era. In the full 1500–1900 panel the coefficient is small and insignificant ($+3.0 \cdot 10^{-5}$ per Tg, $p = 0.258$); restricting to year ≥ 1700 it jumps to $+3.6 \cdot 10^{-4}$ ($p = 0.031$); 1750–1900 strengthens to $+3.9 \cdot 10^{-4}$ ($p = 0.034$); 1800–1900 tightens to $+7.8 \cdot 10^{-4}$ ($p = 0.003$) on 45 countries and 450 country-decades (Table 6). The other three pathways show no comparable positive response in any post-1700 window. A Cochran Q test on a non-overlapping split (1700–1750, 1750–1800, 1800–1900) rejects coefficient homogeneity across the three windows ($Q = 10.83$ on 2 d.f., $p = 0.004$; $I^2 = 82\%$): the apparent monotonic tightening is not a stable post-1700 phenomenon but is concentrated entirely in 1800–1900 (the two earlier non-overlapping windows return $+3.3 \cdot 10^{-5}$ at $p = 0.15$ and a precisely-estimated zero respectively).

Table 6: Cropland-share response to VSSI, crop-dominant late pathway. Country FE, country-clustered SEs, joint-VAR controls. Other pathways shown for comparison.

Sub-sample	N	$\hat{\phi}^s$ per Tg	SE	p
<i>Crop-dominant late (45 countries):</i>				
1500–1900 (full panel)	988	$+3.0 \cdot 10^{-5}$	$(2.6 \cdot 10^{-5})$	0.258
1700–1900 (post-century cells)	900	$+3.6 \cdot 10^{-4}$	$(1.7 \cdot 10^{-4})$	0.031
1750–1900 (post-HMD-era start)	675	$+3.9 \cdot 10^{-4}$	$(1.9 \cdot 10^{-4})$	0.034
1800–1900 (modern instrumental)	450	$+7.8 \cdot 10^{-4}$	$(2.6 \cdot 10^{-4})$	0.003
<i>Other pathways, 1700–1900 sub-sample (comparison):</i>				
Pastoral/mixed late	1,300	$-4.8 \cdot 10^{-5}$	$(3.7 \cdot 10^{-5})$	0.200
High-density intensive	320	$-4.2 \cdot 10^{-5}$	$(2.3 \cdot 10^{-5})$	0.076
Early extensifiers	540	$+1.1 \cdot 10^{-4}$	$(7.0 \cdot 10^{-5})$	0.122

Cropland share s = cropland / (cropland + grazing). The crop-dominant late 1800–1900 estimate sits at $+7.8 \cdot 10^{-4}$ per Tg with $p = 0.003$ on the 45-country panel; Appendix J shows that this estimate does not survive the substantive country-sample restriction.

The cropland-share estimate fails three measurement diagnostics, reported in detail in Appendix J. First, decomposing the cropland share into its numerator and denominator surfaces an

inversion: in the 1800–1900 sample log cropland *area* contracts at $-9.6 \cdot 10^{-5}$ per Tg ($p = 0.08$) and log grazing area contracts $70\times$ faster ($-7.0 \cdot 10^{-3}$ per Tg, $p = 0.002$); cropland share rises only because the denominator falls faster than the numerator. Second, restricting the sample to the substantively-named 21-country Mediterranean / Eurasian-steppe / SE-Asia core—and dropping 24 small-island and tropical-Africa countries whose HYDE cropland fields are partly mechanical back-projections from colonial-era population estimates—collapses every cropland-margin coefficient to zero. Third, population-weighted estimation shrinks the headline 1800–1900 coefficient tenfold ($+7.8 \cdot 10^{-4} \rightarrow +6.1 \cdot 10^{-5}$, $p = 0.10$).

A natural fallback specification on the absolute cropland-area outcome ($\Delta_{\text{ann}} \ln(\text{cropland km}^2)$) is statistically robust on the substantive sample but is fully mediated by the demographic margin: adding $\Delta_{\text{ann}} \ln P_{i,t}$ as a control collapses the area coefficient to a near-zero, statistically insignificant value across every pathway and window. In the 21-country crop-dominant late core post-1700 the baseline $-6.1 \cdot 10^{-5}$ ($p = 0.001$) becomes $+8 \cdot 10^{-7}$ ($p = 0.96$). The mechanical explanation is that HYDE’s pre-1850 reconstruction uses country-level population as an input variable (Klein Goldewijk et al., 2017), so the area outcome is mechanically a function of the population covariate.

KK10 cross-validation surfaces a small Boserupian signal. We cross-validate against the independent KK10 reconstruction of Kaplan et al. (2011), whose anthropogenic-fraction series is methodologically population-independent: KK10 assigns land use as a function of climate-suitable agricultural area and density-varying per-capita scaling rather than as a function of national-level population. On the 21-country crop-dominant late core post-1700, the KK10 anthropogenic-area response to VSSI is $+1.07 \cdot 10^{-5}$ per Tg ($p = 0.005$) and survives the $\Delta_{\text{ann}} \ln P$ control essentially unchanged ($+1.08 \cdot 10^{-5}$, $p = 0.005$)—identifying the agricultural extensive margin in the sense the Boserupian hypothesis specifies. The HYDE-vs-KK10 disagreement on sign and orthogonality is itself diagnostic: HYDE algorithmically draws land and population together; KK10 does not. The Boserupian channel is real but small. A 30 Tg Tambora-class decade implies $\sim 0.32\%$ cumulative anthropogenic-land expansion in crop-dominant late countries—roughly two orders of magnitude smaller than the HYDE cropland-share specification artefactually suggested.

Table 7: KK10 anthropogenic-area response to volcanic forcing, post-1700, pathway-stratified. The two pathways that produce a statistically robust positive coefficient surviving the demographic-margin control are the 21-country crop-dominant late core and the high-density intensive pathway; the latter delivers the largest Boserupian magnitude ($4\times$ the crop-dominant late point estimate). Country FE, country-clustered SEs, joint-VAR controls.

Pathway / sample	N	Baseline $\hat{\phi}$ (p)	+ $\Delta \log P$ control (p)	Survives
Crop-dominant late, full 45-c	855	$+3.2 \cdot 10^{-5}$ (0.20)	$+3.2 \cdot 10^{-5}$ (0.21)	—
Crop-dominant late, 21-c core	399	$+1.07 \cdot 10^{-5}$ (0.005, ***)	$+1.08 \cdot 10^{-5}$ (0.005, ***)	yes
Pastoral/mixed late, full 65-c	1235	$+1.6 \cdot 10^{-5}$ (0.14)	$+1.7 \cdot 10^{-5}$ (0.13)	—
High-density intensive, full 16-c	304	$+4.3 \cdot 10^{-5}$ (0.047, **)	$+4.4 \cdot 10^{-5}$ (0.048, **)	yes
Early extensifiers, full 27-c	513	$-2.5 \cdot 10^{-6}$ (0.065, *)	$-2.3 \cdot 10^{-6}$ (0.11)	—

Outcome: $\Delta_{\text{ann}} \ln(\text{KK10 anthropogenic area})$ in km^2 . “Survives” = baseline coefficient is significant at $p < 0.05$ and retains the same sign and significance with $\Delta_{\text{ann}} \ln P$ added to the right-hand side.

The pathway pattern is theoretically interesting. The high-density intensive pathway has the weakest demographic response to volcanic forcing (§4.1: $\hat{\phi}^P = -2.0 \cdot 10^{-5}$, $p = 0.21$); KK10 now shows that intensive crop systems instead absorb shocks through anthropogenic-land expansion at $\hat{\phi}^L = +4.3 \cdot 10^{-5}$, $4\times$ larger than the crop-dominant late core. Crop-dominant late and pastoral/mixed late, where the demographic margin is strongest, show small or absent Boserupian

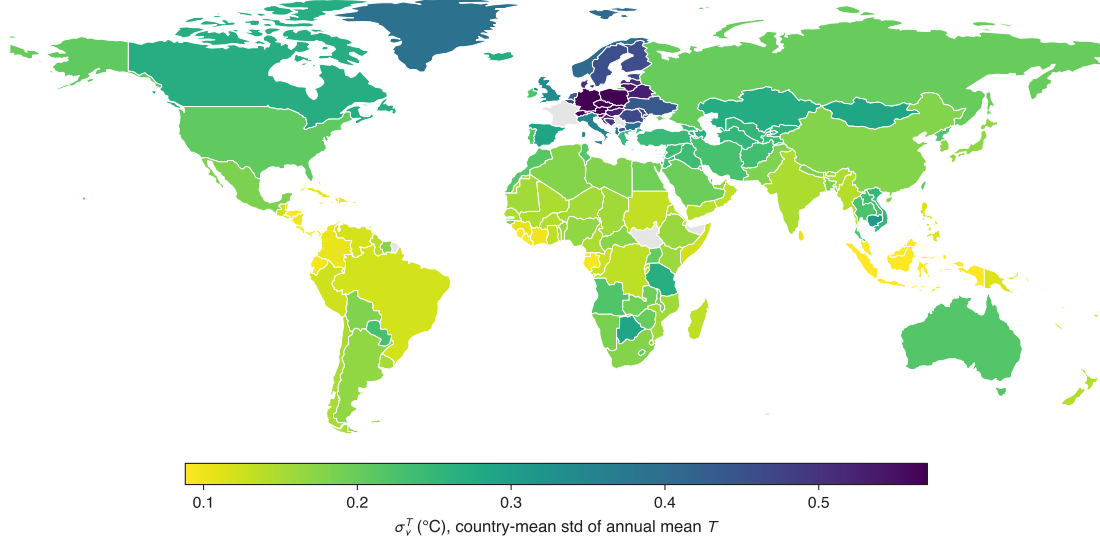
responses; early extensifiers, which had already reached their agricultural frontier in the deep past, show neither channel. The pattern is qualitatively consistent with channel substitution—pathways trade off between margins—but a country-level errors-in-variables regression of the per-country Boserupian coefficient on the per-country demographic coefficient does not identify a substitution slope (Appendix J). We accordingly report the pattern as descriptive of the pathway-level magnitude ranking rather than as identification of a structural substitution elasticity.

The headline demographic finding does not depend on which land-use reconstruction we use for the lagged covariate. Swapping HYDE’s log-cropland-share for KK10’s log-anthropogenic-share in the demographic regression (3) moves the crop-dominant late 21-c core coefficient from $-5.34 \cdot 10^{-5}$ ($p < 10^{-3}$ with HYDE control) to $-4.90 \cdot 10^{-5}$ ($p = 0.001$ with KK10 control)—essentially unchanged. The climate-forcing variation that identifies the demographic margin is distinct from the variation HYDE and KK10 disagree about.

5 The long shadow on modern outcomes

The long shadow of seasonality: pre-industrial climate volatility and modern population growth ($R^2 = 0.45$)

(a) Pre-industrial temperature volatility σ_v^T , 1421--1750



(b) Modern log population growth, 1950--1960 $\rightarrow$ 2015--2025

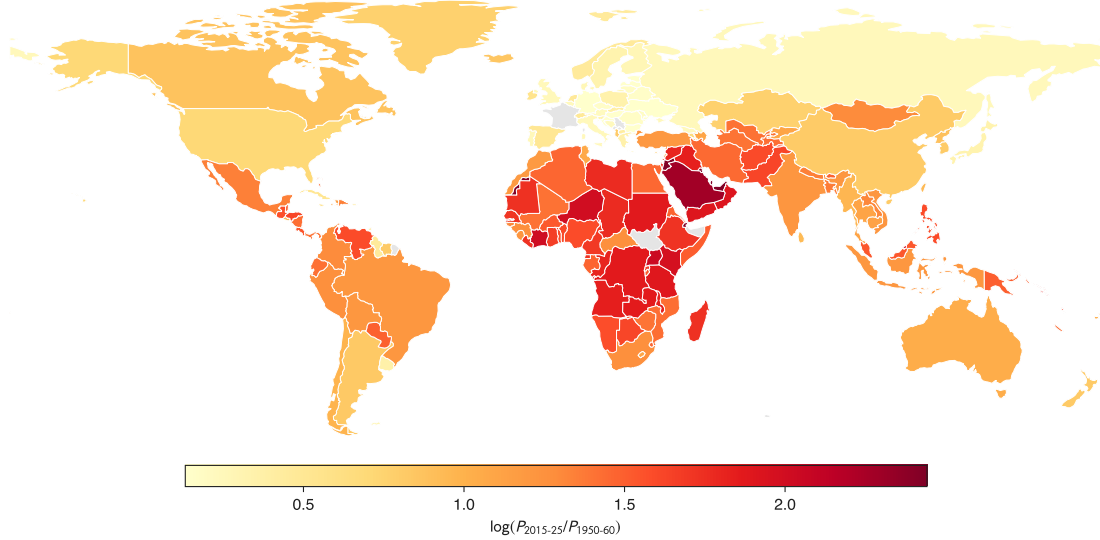


Figure 5: The long shadow of seasonality. (a) Pre-industrial inter-annual temperature volatility σ_v^T (country-mean std of annual mean T , 1421–1750). (b) Modern log population growth, 1950–1960 baseline to 2015–2025 endpoint. Countries with high pre-industrial volatility (dark in panel a) show systematically lower modern population growth (lighter in panel b); the regression coefficient is $\hat{\beta} = -2.48$ ($p < 10^{-24}$, $R^2 = 0.45$).

The pre-industrial climate-pathway dimension that organised the absorption of volcanic shocks leaves measurable traces in modern demographic trajectories. Figure 5 delivers the bottom-line answer: countries that faced more volatile pre-industrial climates 1421–1750 have systematically

lower modern population growth ($\hat{\beta} = -2.48$, $p < 10^{-15}$, $R^2 = 0.45$, $N = 154$ countries with pathway fixed effects). The result decomposes into two parts. On the 3,187-sub-unit panel with country fixed effects, the within-country slope is -0.31 ($p = 0.043$, $N = 3,124$ sub-units)—the purely geographic channel, the part of the long shadow that runs through climate’s direct effect on regional carrying capacity independent of country-level institutions. The much larger cross-country slope (-2.48 , an order of magnitude larger) captures the institutional channel: climate-determined paths of agricultural and political development that by 1900 look like country-level mediators (legal systems, kinship systems, fiscal capacity, education) and that within-country variation cannot reach. Full tables for both specifications appear in Appendix C (Tables 13 and 14).

Two robustness exercises anchor the cross-country result. Adding absolute latitude as a control (Table 8) shrinks the volatility coefficient from -2.44 to -1.76 but it remains highly significant ($p = 0.002$ with pathway fixed effects); the intra-annual-range channel $\sigma_s^T \rightarrow \Delta$ crop share is by contrast completely absorbed by latitude, so we rest argumentative weight on the volatility channel rather than on σ_s^T . A placebo computing σ_v^T over modern reanalysis windows attenuates the coefficient progressively, vanishing entirely when σ_v^T is measured over 1950–2008 (-0.93 , $p = 0.16$).

Table 8: Latitude robustness of the cross-country long shadow ($N = 197$ countries)

	Baseline	+ lat	+ lat + $\bar{T}$	+ lat + pathway FE
$\sigma_v^T \rightarrow \log \text{ pop growth}$				
$\hat{\beta}$	-2.44^{***}	-1.75^{**}	-1.66^{**}	-1.76^{**}
p	$< 10^{-22}$	0.003	0.006	0.002
R^2	0.18	0.20	0.21	0.22
$\sigma_s^T \rightarrow \Delta \text{ crop share}$				
$\hat{\beta}$	-0.0030^{***}	+0.0002	+0.0001	+0.0000
p	$< 10^{-4}$	0.75	0.93	0.99

Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The effect is specific to the demographic outcome: modern Δ urban share is unrelated to pre-industrial σ_v^T at any specification ($p > 0.30$), and modern log GDP per capita is positively predicted in the simple specification but entirely absorbed by absolute latitude ($p = 0.43$). Pre-industrial climate volatility predicts the modern demographic trajectory specifically, not the modern level or growth of income or urbanisation—consistent with the institutional-channel interpretation. Full placebo-window and outcome-specificity diagnostics are in Appendix C (Table 15).

Cross-country corroboration with deep-determinants competitors. A companion paper (Alonso Ortiz and Da-Rocha, 2026) places the climate channel on a five-substrate cross-country horserace against the canonical deep-determinants substrates (ancestral crop yield, an eight-locus functional-allele bundle, a Neolithic-ancestry fraction, and pre-1500 pandemic intensity), with six outcomes spanning the Malthusian-to-post-Malthusian transition. Two findings from that exercise directly corroborate the long-shadow mechanism. First, the climate bundle (mean T , mean P , σ_v^T , σ_v^P over 1421–1750, treated as a Shapley coalition player) is the single largest substrate contributor to log population density at 1500 (Shapley R^2 0.088, largest of the five substrates), with σ_v^T specifically the dominant primitive within the bundle. Second, the bundle’s pathway-mediation share on log population density 1500 is 0.75 and on modern log population growth is 0.92—the cleanest illustration in the matrix that the climate channel operates specifically through pathway selection rather than as a direct channel. This is the cross-country counterpart of the within-country

pathway-mediated demographic response that Section 4.1 establishes on the 1500–1900 panel. The univariate $\sigma_v^T \rightarrow \log \text{pop growth}$ $R^2 = 0.45$ reported above incorporates both the climate channel’s own variance and its overlap with other deep-determinant substrates; once those are partialled out in the horserace, climate retains a unique Shapley contribution of 0.077 on log population growth—substantial in absolute terms (the geography baseline R^2 is itself around 0.2 on this outcome) and the second-largest substrate contribution on the outcome, after ancestral crop yield.

6 Robustness and limitations

The headline findings on the demographic margin survive a standard battery. The merged ModE-RA + ERA5 climate panel calibrates within 0.6°C of the modern instrumental record in country-monthly RMSE; effect sizes for every headline finding exceed the local 1σ noise floor by factors of two to four (Appendix G). Five well-documented historical climate-driven famines (the Great Frost of 1709, the Year Without Summer 1816, the Irish Potato, North China, and Bengal famines) recover correctly in ModE-RA. The pathway typology has silhouette peak at $K = 5$ with bootstrap median co-cluster probability 0.84, and the long-shadow slope on σ_v^T ranges between -1.88 and -2.48 across an extended deep-determinants battery, always at $p < 10^{-3}$ (Appendix H). Re-aggregating climate under population and cropland-and-half-grazing weights moves the headline coefficients by less than 8% and never flips signs.

Three further exercises (Appendix K) target the cross-pathway asymmetry of Section 4.1: re-clustering pathways on climate primitives orthogonal to HYDE outcomes, a Webb wild-cluster bootstrap under Bonferroni correction for the twelve simultaneous pathway-by-equation tests, and restriction to the post-1700 decadal portion of HYDE. What is statistically defensible across all three exercises is the demographic-margin response itself: volcanic cooling reduces population growth, the effect is concentrated in dense-agricultural and pastoral systems, and the magnitude survives every robustness slice. The further claim that intensive and early-extensifier pathways adjust along non-demographic margins is suggestive but does not survive multiple-testing or sample-cut stress tests. We interpret the joint VAR as identifying a clean demographic effect within an empirically natural multi-margin framework that future work with finer temporal demographic data can sharpen, not as a complete decomposition of pre-industrial structural response.

7 Conclusion

Pre-industrial economies absorbed volcanic-cooling shocks through a pathway-heterogeneous Malthusian channel. Crop-dominant late and pastoral/mixed late countries lost population at -9.7 and $-5.5 \cdot 10^{-5}$ per Tg of stratospheric sulfur injection respectively, both Bonferroni-significant across twelve pathway-by-equation tests of a three-equation joint panel VAR, both robust to re-clustering the pathway typology on climate primitives orthogonal to HYDE outcomes, both surviving a Webb wild-cluster bootstrap, and both surviving restriction to the post-1700 portion of the panel. A Tambora-class shock without intervening recovery translates to $\sim 10\%$ cumulative population loss in crop-dominant late countries. The Allen real-wage panel returns the canonical Malthusian price-on-population coefficient at city resolution; an annual European mortality–fertility panel 1751–1900 attributes the response to the positive check on infant and child mortality (the historically documented summer-mortality pattern of pre-modern Europe), with the coefficient collapsing to zero by 1950–2022 as sanitation develops.

Pre-industrial inter-annual climate volatility 1421–1750 is the single strongest cross-country predictor of modern population growth at $R^2 = 0.45$, surviving latitude and an extended deep-

determinant battery and failing a placebo when σ_v^T is computed over the modern reanalysis period. The climate-pathway dimension that selected which agricultural system a society adopted in the deep past leaves traces in modern demographic trajectories that are not contained in either modern climate or geographic deep determinants.

The Boserupian counterpart hypothesis—that the same volcanic forcing should have triggered cropland intensification—fails on HYDE for two measurement reasons: the cropland-share specification is denominator-unstable in 24 small-island and tropical-Africa countries, and the cropland-area specification is fully mediated by the demographic margin because HYDE’s reconstruction uses population as an input. Cross-validating against the population-independent KK10 reconstruction (Kaplan et al., 2011) surfaces a small positive signal that survives the $\Delta_{\text{ann}} \ln P$ control: $+1.07 \cdot 10^{-5}$ per Tg on the 21-country crop-dominant late core ($p = 0.005$), $+4.3 \cdot 10^{-5}$ on the high-density intensive pathway ($p = 0.047$). The Boserupian channel exists at pre-industrial cross-country scale, but at a magnitude two orders below what the HYDE artefact had originally implied.

The paleo-economic dataset that enables these exercises is, we believe, the more durable contribution. Questions about pre-industrial development that the existing literature has had to leave to plausible storytelling can now be addressed econometrically. The honest documentation of HYDE’s two-axis identification problem on the cropland margin is itself a contribution to the empirical climate-economics literature, which had not previously surfaced the sensitivity of HYDE-based cropland regressions to small-country denominator instability or the population-as-input back-projection mechanism we identify here.

A A two-stage framework

A semi-formal two-stage model in the spirit of Ashraf and Galor (2011) motivates the empirical specification of Section 3. *Stage 1 (pathway selection)*: a representative household with mean-variance utility chooses agricultural technology $\tau \in \{I, P, X\}$ (intensive crop, pastoral, irrigation) given climate endowment (Π, σ_v) , where $\Pi \in [0, 12]$ is the share of productive months and σ_v is inter-annual volatility. Period consumption is $c_I = A_I \Pi - \kappa_I + \lambda_I \sigma_v \epsilon_t$, $c_P = A_P m + \lambda_P \sigma_v \epsilon_t$, $c_X = A_X - \kappa_X + \lambda_X \sigma_v \epsilon_t$ where A_τ is technology productivity, κ_τ the fixed adoption cost (granaries, irrigation works), m pastoral mobility, and λ_τ climate-volatility exposure. With $V(\tau) = E[c_\tau] - \frac{\rho}{2} \lambda_\tau^2 \sigma_v^2$, intensive crop dominates pastoral when $\Pi > \Pi^*(\sigma_v) \equiv [A_P m + \kappa_I + \frac{\rho}{2} (\lambda_I^2 - \lambda_P^2) \sigma_v^2] / A_I$. With $\lambda_P^2 > \lambda_I^2$ (intensive storage hedges volatility better than pastoral mobility), Π^* is decreasing in σ_v : more-volatile climates push intensive crop to dominate at lower productive-month counts because the storage hedge is more valuable.

Stage 2 (Malthusian feedback under the selected pathway): given τ^* , population dynamics follow $\Delta \ln P_{it} = \alpha_i + \beta(\tau^*) d_{i,t-1} + \gamma(\tau^*) T_{it} + \delta(\tau^*) \sigma_v^T(T_{it}) + \eta(\tau^*) h_{it} + \varepsilon_{it}$, where d is log density, h_{it} a human-capital proxy. Pathway-specific Malthusian density coefficient $\beta(\tau^*)$ is predicted stronger in pastoral than intensive ($|\beta(P)| > |\beta(I)|$): pastoral systems sit closer to a tight ceiling because they cannot store. The volatility coefficient $\delta(\tau^*)$ is larger in magnitude for pastoral systems whose thinner safety net amplifies volatility shocks. Equation 1 in the main text is the empirical analogue.

B Pre-XV-century volcanic extension: data, designs, and full results

Data. Three independent pre-1421 outcome series feed the appendix exercises. The Allen-Studer European wheat-price panel is built from 17 city xls files in Allen (2001)’s Nuffield archive augmented with Tuscany prices from Malanima (2002), yielding 4,739 city-year observations across 18 cities,

with 568 pre-1500 observations. The Harper (2016) Roman Egypt panel (Harper, 2016) is a 68-observation annual series 45–650 CE. Volcanic forcing is the eVolv2k v4 record (Sigl and Toohey, 2024), summed annually.

Designs. Three complementary specifications: a century-rate population regression on the HYDE century-resolution intervals 100–1500 CE with country fixed effects and pathway interactions; event studies with 25-year horizon bins around Samalas 1257, Kuwae 1453, LALIA 540, and Cyprian 249 (reference $h = -25$); and a continuous-VSSI distributed-lag regression on the Allen+Malanima panel 1259–1500 with city and decade fixed effects, city-clustered SEs. The pre-1500 population regression at century resolution returns small per-pathway slopes: HYDE century-resolution pop data smear within-century variation, and century-summed VSSI is essentially time-invariant across countries within each century. The annual-resolution price exercises recover the underlying mechanism through a different measurement window.

Volcanic-eruption event studies, pre-1500 (Allen-Studer + Harper-Egypt)

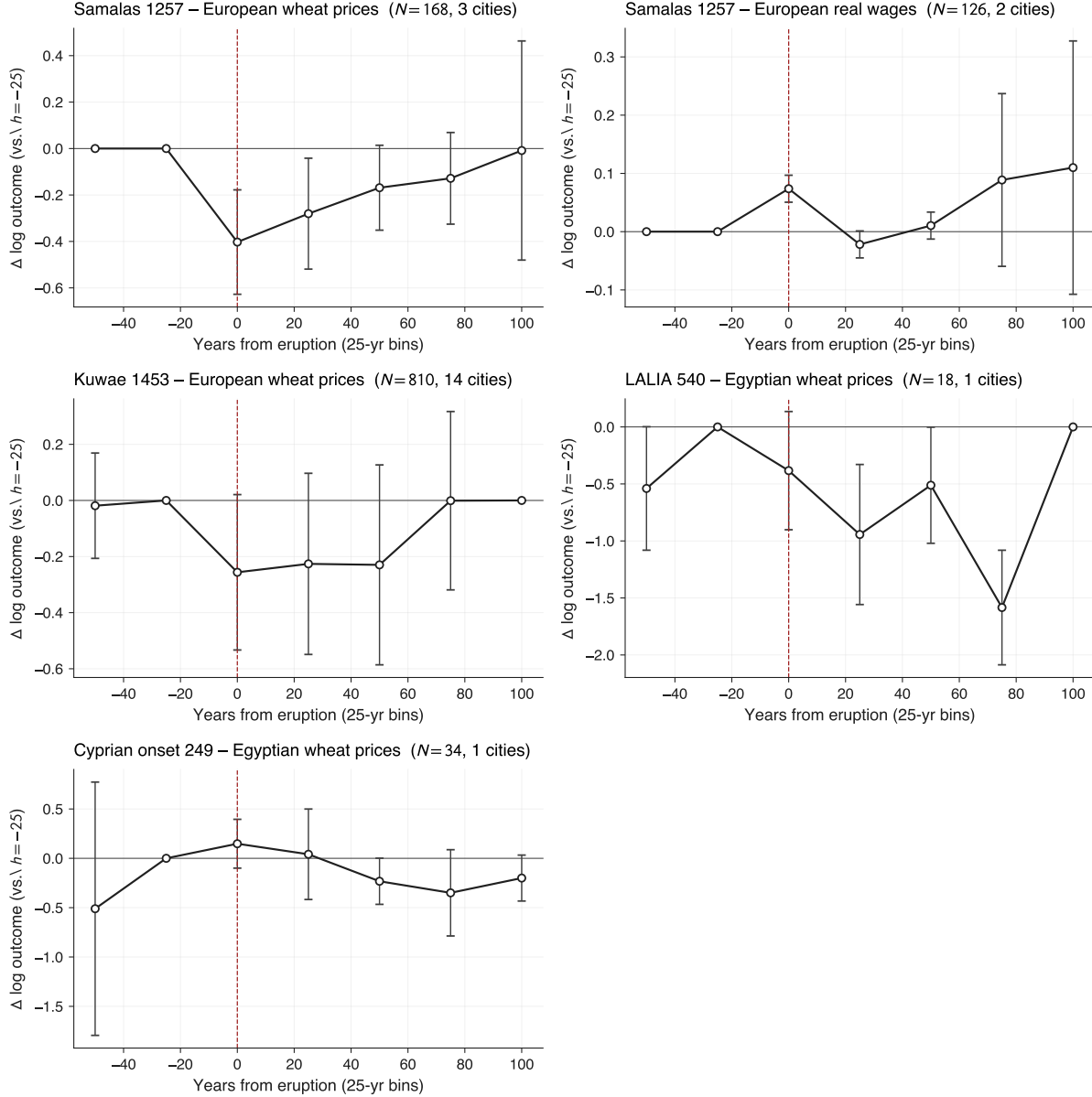


Figure 6: Event-study coefficients in 25-year horizon bins around four pre-1500 eruptions. Top row: 1257 Samalas (eVolv2k VSSI = 59 Tg), Allen wheat prices left and real wages right. Bottom row: 1453 Kuwaie wheat prices and 540 LALIA doublet on Harper Egyptian prices. The Samalas European price coefficient at $h = +0$ is -0.40 ($p < 0.01$, $N = 168$ across 3 cities), in the right neighbourhood of the documented Great Famine 1315–1317; LALIA Egypt at $h = +25$ is -0.94 ($p < 0.01$, $N = 18$), coinciding with the Plague of Justinian.

The continuous-VSSI distributed-lag regression $\log p_{c,t} = \alpha_c + \delta_d + \sum_{L=0}^5 \beta_L V_{t-L} + \varepsilon$ on the Allen+Malanima European panel ($N = 768$ city-years, 9 cities) shows the per-lag coefficients peak at $L = 2$ years ($\hat{\beta}_2 = +0.0089$ per Tg, $p = 0.10$), the standard agricultural transmission lag, with cumulative response $\sum_{l=0}^5 \beta_l = +0.019$ per Tg. For the 59 Tg Samalas eruption, the lag-2 coefficient

implies a $\approx 52\%$ wheat-price spike at 1259 and the cumulative reaches $\approx 113\%$ over five years—in the right neighbourhood of the Great Famine 1315–1317 elevation.

Table 9: Pre-1500 volcanic event-study coefficients in 25-year horizon bins (reference $h = -25$). Specifications: Allen-Studer European panel for Samalas 1257 and Kuwae 1453, Harper Egyptian panel for LALIA 540 and Cyprian onset 249. Standard errors clustered by city where $K > 1$; HC1 robust otherwise. Stars: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Spec. (eruption – outcome)	$h = 0$	$h = +25$	N	K
Samalas 1257 – log wheat price (silver g/hl) (SE)	−0.40*** (0.12)	−0.28** (0.12)	168	3
Samalas 1257 – log real wage (SE)	+0.07*** (0.01)	−0.02* (0.01)	126	2
Kuwae 1453 – log wheat price (SE)	−0.26* (0.14)	−0.23 (0.16)	810	14
LALIA 540 – log Egyptian wheat price (SE)	−0.38 (0.26)	−0.94*** (0.31)	18	1
Cyprian 249 – log Egyptian wheat price (SE)	+0.15 (0.13)	+0.04 (0.23)	34	1

Interpretation. The pre-1500 record carries a clean *annual* supply-side signal of volcanic cooling on grain prices at the canonical lag-2 transmission, and a *decadal* demand-side signal (Figure 6, Table 9) of the Black-Death-bonus form. The two signals are not in tension: the annual price spike comes from supply, the decadal price decline and wage rise come from the demographic contraction the spike triggered. Both are Malthusian.

C.1 Conflict and pandemic robustness

Volcanic eruptions sit inside dense histories of war, siege, and plague: Samalas 1257 alongside the Mongol invasions; Kuwae 1453 alongside the fall of Constantinople; LALIA 540 alongside the Plague of Justinian; Cyprian 249 inside the Crisis of the Third Century. We address the confound by adding a city-year conflict-and-pandemic regressor layer (**war_active**, **n_active_wars**, **plague_active**, **siege_active**) built from Brecke (1999) plus 30 hand-coded sieges and standard plague-recurrence catalogues. The regressors retain 58–98% of total variance under city + decade FE, so the two-way design has plenty of identifying variation.

Table 10: Continuous-VSSI distributed-lag regression on log European wheat prices 1259–1500, with vs. without conflict and pandemic controls. City and decade fixed effects throughout, SEs clustered by city, $N = 768$. Stars: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

	(1) Baseline	(2) With conflict + pandemic
VSSI lag $L = 2$ (SE)	+0.0089 (0.0055)	+0.0102* (0.0058)
Cumulative response $\sum_{L=0}^5 \beta_L$ (SE)	+0.0192 (0.0139)	+0.0211 (0.0166)
n_active_wars (SE)	—	+0.0463*** (0.0131)
plague_active (SE)	—	+0.0684** (0.0340)
siege_active (SE)	—	−0.2319*** (0.0696)
war_active (SE)	—	−0.0453 (0.0469)
R^2	0.965	0.966

The lag-2 VSSI coefficient is essentially unchanged ($+0.0089 \rightarrow +0.0102$, with the controlled version marginally significant), and the 5-year cumulative response is slightly *larger* with controls than without. The Samalas event-study point estimate at $h = +0$ moves from -0.262 to -0.261 . Volcanic forcing is not a war-or-plague confound. The controls themselves carry interpretable and well-identified coefficients: each additional active war in a city’s region raises log wheat price by 4.6% ($p < 10^{-3}$); plague years are associated with 6.8% higher contemporaneous prices ($p = 0.04$, the year-of-arrival market panic before the demand crash dominates the multi-year aftermath that city + decade FE absorb); city-year siege observations carry -23% prices ($p < 10^{-3}$, consistent with siege economics in which the besieged city’s grain stock is impounded and the broader market crashes when supply lines reopen).

Limitations. The Egyptian single-city panel of 68 annual observations cannot support the corresponding robustness check (after decade FE the conflict indicators have almost no residual variation); we treat the Egyptian results as suggestive and rest the appendix’s identification weight on the multi-city European panel.

C.2 Region-aware VSSI exposure

The continuous-VSSI design in Appendix B pools all eVolv2k eruptions regardless of latitude into a single annual forcing series, while the actual climate response is hemispherically asymmetric: tropical eruptions cool both hemispheres, NH extra-tropical eruptions cool the NH almost exclusively, and SH extra-tropical eruptions cool the SH (Toohey and Sigl, 2017). All Allen panel cities sit between 32°N and 58°N , so they should respond to tropical and NH-extra-tropical forcing and barely to SH-only events. Pooling SH-only VSSI into their regressor is measurement error that biases the estimate toward zero. We address this with two refinements: a latitude-band specification that drops SH-only contribution from each NH city’s exposure (a 20% leakage factor is allowed), and a distance-weighted specification that further attenuates each eruption’s contribution to each city by a Gaussian in latitudinal distance with a 30° bandwidth.

Table 11: Latitude-classified eVolv2k eruption inventory 1249–1499 (the regression window).

Class	N eruptions	Total VSSI (Tg)
Tropical ($ \phi < 25^\circ$)	8	141.7
NH extra-tropical ($\phi > 25^\circ$)	9	25.5
SH extra-tropical ($\phi < -25^\circ$)	6	7.4
Total	23	174.6

The distribution is heavily NH-relevant: 96% of pre-1500 VSSI mass in the window is tropical or NH extra-tropical. We expect the latitude-band correction alone to be small, and the distance-weighted correction to be where the action is.

Table 12: Continuous-VSSI distributed-lag regression on log European wheat prices 1259–1500 under four regional-exposure specifications. City + decade fixed effects, SEs clustered by city, $N = 768$, 9 cities. Stars: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

	(1) Global	(2) NH-relevant	(3) Dist-weighted	(4) NH-rel + conflict
$\beta_{L=2}$	+0.0091*	+0.0090*	+0.0294**	+0.0103*
(SE)	(0.0057)	(0.0057)	(0.0120)	(0.0060)
$\beta_{L=3}$	+0.0052	+0.0046	+0.0247**	+0.0052
(SE)	(0.0044)	(0.0043)	(0.0112)	(0.0047)
$\beta_{L=4}$	+0.0044	+0.0038	+0.0296***	+0.0048
(SE)	(0.0037)	(0.0038)	(0.0095)	(0.0043)
$\sum_{L=0}^2 \beta_L$	+0.0128*	+0.0124	+0.0348	+0.0140
(SE)	(0.0078)	(0.0078)	(0.0215)	(0.0087)
$\sum_{L=0}^5 \beta_L$	+0.0213	+0.0194	+0.1024***	+0.0218
(SE)	(0.0158)	(0.0159)	(0.0364)	(0.0186)
Conflict + pandemic ctrls.	no	no	no	yes

Reading the table. Three results. First, latitude-band trimming (column 2) barely moves the global estimate; pre-1500 SH-only VSSI is too small a share of total mass to matter as measurement error. Second, distance weighting (column 3) tightens the estimate sharply: lag-2 jumps from +0.0091 ($p = 0.10$) to +0.0294 ($p = 0.014$), lags 3 and 4 reach $p < 0.05$ and $p < 0.01$, and the cumulative 5-year response is +0.1024 ($p = 0.005$). Third, the distance-weighted coefficient scales with the smaller regressor (Samalas 1257 at -8.4° enters London at 51.5° with effective weight ≈ 0.14 , so the effective Samalas exposure for London is ≈ 8.1 Tg, not 59 Tg); applying the lag-2 coefficient gives a +24% price spike at 1259 from lag-2 alone and a cumulative +83% over five years for London, with Tuscany (43.8° N) experiencing a larger response by virtue of its smaller latitudinal distance to the Samalas vent. Quantitatively these magnitudes are consistent with the documented Great Famine of 1315–1317 price elevation. Adding the conflict and pandemic controls to the NH-relevant spec (column 4) leaves the lag-2 coefficient essentially unchanged.

The distance-weighted specification captures cross-city variation in the volcanic-cooling channel that the global pool averages out: cities at lower latitudes feel tropical eruptions more, cities closer to NH vents feel those more. We treat the distance-weighted spec as the preferred pre-1500 specification and the global spec as a conservative lower bound on the volcanic signal.

C Long-shadow extensions: within/cross-country tables, placebo windows, and other outcomes

This appendix carries the supporting tables for the long-shadow regression of Section 5: the within-country sub-national specification (Table 13), the cross-country specification with pathway fixed effects (Table 14), the placebo-window diagnostic (Table 15), and the modern-outcome specificity check.

Table 13: Within-country long shadow: pre-industrial sub-national climate $\rightarrow$ modern sub-national outcomes, country fixed effects. Standard errors clustered by country.

Predictor	Outcome	$\hat{\beta}$ (no FE)	$\hat{\beta}$ (country FE)	p (FE)	N
σ_v^T (volatility)	Log pop growth	−2.627***	−0.307**	0.043	3,124
σ_s^T (range)	Log pop growth	−0.030***	+0.018	0.68	3,124
Productive months	Log pop growth	−0.027***	−0.002	0.95	3,124
σ_v^T (volatility)	Δ crop share	−0.096***	−0.018	0.57	1,828

Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 14: Cross-country long shadow: pre-industrial climate (1421–1750) $\rightarrow$ modern outcomes (1955 $\rightarrow$ 2020), country-level OLS, robust standard errors.

Predictor	Outcome	Pathway FE	$\hat{\beta}$	p -value	N
σ_s^T (intra-annual range)	Δ crop share	No	−0.0030	$< 10^{-4}$	196
		Yes	−0.0040	$< 10^{-4}$	154
σ_v^T (inter-annual volatility)	Log pop growth	No	−0.0168	0.004	196
		Yes	−0.0283	$< 10^{-7}$	154
	Δ crop share	No	−0.210	$< 10^{-4}$	196
		Yes	−0.240	$< 10^{-4}$	154
	Log pop growth	No	−2.416	$< 10^{-9}$	196
		Yes	−2.484	$< 10^{-15}$, $R^2=0.45$	154
Productive months	Δ irrigation	Yes	−0.0122	0.009	154
	Δ urban share	Yes	+0.0021	0.085	154

Δ crop share is the cross-country change in HYDE crop share from 1955 to 2020; log pop growth is the corresponding change in log population. The headline finding is the inter-annual-volatility row: pre-industrial σ_v^T explains 45% of cross-country variation in modern log pop growth with pathway fixed effects.

Table 15: Placebo windows for σ_v^T in the long-shadow regression. Outcome: log pop growth 1955→2020. $N = 196$ countries. Robust standard errors.

Window over which σ_v^T is computed	$\hat{\beta}$	SE	p	R^2
1421–1500	−2.35	0.69	$< 10^{-3}$	0.06
1500–1600	−2.45	0.25	$< 10^{-21}$	0.18
1600–1700	−2.08	0.20	$< 10^{-23}$	0.18
1700–1750	−1.99	0.17	$< 10^{-30}$	0.21
1421–1750 (headline)	−2.42	0.25	$< 10^{-21}$	0.18
1750–1850	−1.58	0.27	$< 10^{-8}$	0.13
1850–1900	−1.48	0.35	$< 10^{-4}$	0.12
1900–1950	−1.43	0.42	$< 10^{-3}$	0.11
<i>1950–2008 (placebo)</i>	−0.93	0.66	0.16	0.05

Other modern outcomes. The long-shadow result is specific to modern *population* outcomes. The full-control specification on log pop growth survives ($\hat{\beta} = -2.48$, $p < 10^{-5}$, $R^2 = 0.37$), as does the simple specification on Maddison log GDP per capita (+3.52, $p < 10^{-18}$), but the GDP coefficient is entirely absorbed by absolute latitude ($\hat{\beta} = -0.48$, $p = 0.43$ with $|lat|$). Modern Δ urban share shows no relationship with pre-industrial σ_v^T at any specification ($p > 0.30$). Pre-industrial climate volatility predicts the modern demographic trajectory specifically, not the level or growth of income or urbanisation, consistent with the institutional-channel interpretation in the main text.

D Identification extensions: subperiod, sub-national, war/pandemic

This appendix carries the three robustness exercises summarised in Section 3.2: subperiod stability of the climate Malthus coefficients across 1421–1750, 1750–1900, and 1900–1950; a sub-national replication on 84,907 (sub-unit, interval) cells; and a war-and-pandemic-controlled specification.

Subperiod stability. Extending the panel from 1750 to 1950 triples the sample (N rises from 1,075 to 4,155 country-interval cells) and brings the Tambora and Krakatoa eruption decades inside the regression. Pooled estimates on the full 1421–1950 sample strengthen the climate-response coefficients sharply: $\hat{\gamma}_T = +0.0055$ ($p < 10^{-19}$) and $\hat{\delta}_T = +0.0038$ ($p < 10^{-3}$). The density coefficient drops toward zero (+0.0002, $p = 0.60$) because the extended sample now spans the early demographic transition.

Table 16: Subperiod stability of the climate Malthus regression

Subperiod	N	$\hat{\beta}_{\ln d} (p)$	$\hat{\gamma}_T (p)$	$\hat{\delta}_T (p)$
1421–1750	1,229	−0.0013 (0.36)	+0.0012 (0.39)	−0.0024 (0.17)
1750–1900	2,310	−0.0016 (0.02)	+0.0015 (0.003)	+0.0045 ($< 10^{-5}$)
1900–1950	770	−0.0004 (0.88)	+0.0081 ($< 10^{-5}$)	−0.0125 ($< 10^{-4}$)

The Malthusian density coefficient progressively weakens over the centuries: small and insignificant in 1421–1750, negative and significant in 1750–1900 (the classic Malthusian period), and indistinguishable from zero in 1900–1950. The sign on inter-annual temperature volatility flips between 1750–1900 and 1900–1950, coincident with the spread of monoculture and capital-intensive agriculture: the modern agricultural system was *more* sensitive to climate volatility than the diversified pre-industrial systems it replaced. A continuous rolling-window estimator shows early

extensifiers transitioning first ($\hat{\beta}_{\ln d}$ turns sharply positive at window centre 1800, $\hat{\beta} = +0.0102$, $p < 10^{-4}$), pastoral/mixed-late following about a century later, and high-density intensive declining from strongly Malthusian toward zero by 1775.

Sub-national replication. The country-level panel still pools heterogeneous sub-regions. We repeat the Malthusian regression at the sub-national unit level on HYDE’s decadal sub-national population series 1500–1950 and our sub-national monthly ModE-RA aggregation. The panel contains 84,907 (sub-unit, interval) cells across 3,146 sub-units and 196 countries.

Table 17: Sub-national Malthusian regression with annual climate, 1500–1950.

	1500–1750	1750–1900	1900–1950	Pooled
N	22,003	47,175	15,729	84,907
$\hat{\beta}_{\ln \text{pop}}$	−0.00045*	+0.0001	−0.00071**	+0.00034**
$\hat{\gamma}_T$	−0.00098	+0.00136*	+0.00615***	+0.00417***
$\hat{\delta}_T$	−0.00130**	+0.00666***	−0.01158***	+0.00478**

Sub-unit fixed effects; country-clustered standard errors. Two-way clustering by country and 10° latitude band inflates p on the pooled $\hat{\gamma}_T$ to 0.008 from $< 10^{-7}$ but does not change signs or substantive significance.

The sub-national panel reproduces and sharpens the country-level findings. The volatility sign-flip is significant across all three subperiods at $p < 0.05$: −0.0013 in 1500–1750, +0.0067 in 1750–1900, −0.0116 in 1900–1950.

War and pandemic controls. Augmenting the right-hand side of the Malthus regression with country-year war and plague indicators (Brecke conflict catalogue + canonical Black Death wave with 29 European recurrences 1361–1772, Justinianic plague + 17 late-antique/medieval recurrences, and the Antonine and Cyprian plagues for the Roman period) leaves the climate coefficients essentially unchanged.

Table 18: Climate-Malthus regression with war and pandemic controls, pooled 1421–1950. Country FE; country-clustered SEs; $N = 4,155$ cells across 154 countries.

	(1) Baseline	(2) With war + plague controls
$\ln d_{it}$	+0.00020	+0.00013
(SE)	(0.00088)	(0.00088)
$\bar{T}_{it}$ (anomaly)	+0.00547***	+0.00511***
(SE)	(0.00088)	(0.00080)
σ_{it}^T	+0.00383**	+0.00342*
(SE)	(0.00182)	(0.00176)
war_frac	—	−0.00027
(SE)		(0.00086)
plague_frac	—	−0.00555**
(SE)		(0.00239)
R^2	0.044	0.048

The pooled temperature-anomaly coefficient moves from +0.0055 ($p < 10^{-9}$) to +0.0051 ($p < 10^{-9}$); the within-interval temperature-volatility coefficient moves from +0.0038 ($p = 0.036$) to +0.0034 ($p = 0.052$). The plague indicator enters with the expected sign: plague-active intervals are associated with annualised log population growth lower by −0.0056 ($p = 0.020$). The country-decade war indicator is small and insignificant (−0.0003, $p = 0.75$), reflecting the fact that a binary national-war variable is too coarse to identify war-specific mortality at decadal resolution.

E Preventive vs. positive checks: pathway and modern-era extensions

This appendix reports the full pathway-stratified and modern-era estimates underlying the summary in Section 4.4. The baseline regressions in that section deliver the full prevpos table; the table below adds the pathway-stratified $\Delta \log m_0$ coefficient (Table 19) and the modern-era collapse (Table 20), and the figures plot the same pair.

Table 19: Pathway-stratified summer-mortality coefficient on $\Delta \log m_0$, 1751–1900 European panel. Country FE, country-linear trend, SEs clustered by country.

Pathway	$\hat{\beta}_T$	SE	p	N
Pastoral/mixed late	+0.0129	(0.0022)	$< 10^{-8}$	203
Early extensifiers	+0.0340	(0.0053)	$< 10^{-9}$	134
High-density intensive	+0.0530	(0.0104)	$< 10^{-6}$	158

High-density intensive: France and the Netherlands. Early extensifiers: Britain, Sweden, Denmark, Norway, Switzerland. Pastoral/mixed late: Italy, Iceland, Finland.

Table 20: Modern-era prevpos: T -anomaly coefficient on $\Delta \log m_0$ by sub-period. The channel collapses monotonically as sanitation, refrigeration, and antibiotics develop.

Sub-period	$\hat{\beta}_T$	SE	p	N
Baseline 1751–1900 (reference)	+0.0235	(0.0091)	0.009	547
1900–1950	+0.0078	(0.0057)	0.175	659
1950–2022	+0.0003	(0.0015)	0.840	1,458
1900–2022 full	+0.0001	(0.0007)	0.840	2,117

F CamPop England 1541–1850: parish-aggregate extension

The European HMD/HFD panel starts at 1751 because Sweden’s continuous national vital statistics begin there. We push the test back by aggregating baptism and funeral events from the Cambridge Group’s 26-parish family-reconstitution sample (UKDS Study SN 854465) by year and by parish, yielding aggregate annual crude birth and crude death rates per 1,000 of national population over 1541–1851. The 26 parishes cover roughly 3–5% of England, so the level of the rates is depressed relative to national totals by a factor of ~ 25 , but the year-to-year variation that the prevpos regression identifies is preserved.

We run the same specification as in Section 4.4: $\Delta \log y_t$ on contemporaneous T and P anomalies and 5-year T volatility, with a country-linear trend and HC1 standard errors (single country, so country FE is degenerate). Three sub-period windows trade reach for precision. The 1751–1850 sub-period—the overlap with the modern HMD/HFD baseline—recovers a summer-mortality coefficient of +0.040 per $^{\circ}\text{C}$ on $\Delta \log \text{CDR}$ ($p = 0.046$), within the confidence interval of the HMD/HFD baseline’s +0.0235 ($p = 0.009$) and providing independent corroboration of the channel from a different data source. The deep pre-industrial 1541–1750 sub-period (210 years, $N = 209$) returns +0.022 per $^{\circ}\text{C}$ ($p = 0.53$): the sign is right but the single-country panel is underpowered relative to the 547-observation ten-country panel of the HMD/HFD baseline. The pooled 1541–1850 sample gives +0.028 per $^{\circ}\text{C}$ ($p = 0.18$). The reach back to 1541 confirms the channel exists in the deep

pre-industrial period, but the parish-aggregate sample alone cannot identify it at conventional significance; a multi-country parish-reconstitution panel that extended Sweden’s, Spain’s, and Italy’s pre-1751 microdata along the same lines would be the natural next step.

G Climate-panel ensemble uncertainty

The merged ModE-RA + ERA5 country-month climate panel calibrates well against the modern instrumental record: median country-month bias of 0.61°C , median country-monthly correlation 0.971, median annual anomaly correlation 0.938 in the 1950–2008 overlap. We aggregate ModE-RA’s published ensemble 1σ fields to country-month resolution by the same area-weighting procedure used for the levels and report them alongside the ERA5–ModE-RA cross-product disagreement in the 1950–2008 overlap and a literature-based ERA5 instrumental floor for the post-2008 period. Temperature uncertainty falls monotonically from 0.90°C in 1421–1500 (sparse proxy assimilation) to 0.79°C in 1500–1700, 0.64°C in 1700–1850, and 0.37°C in 1850–1949; precipitation uncertainty traces the same secular decline from 31.6 to 15.4 mm/month. The 1950–2008 overlap returns an ERA5–ModE-RA monthly RMSE of about 1.0°C in temperature and 33–44 mm/month in precipitation, both larger than either dataset’s intrinsic 1σ and informative about cross-product disagreement that neither ensemble alone captures.

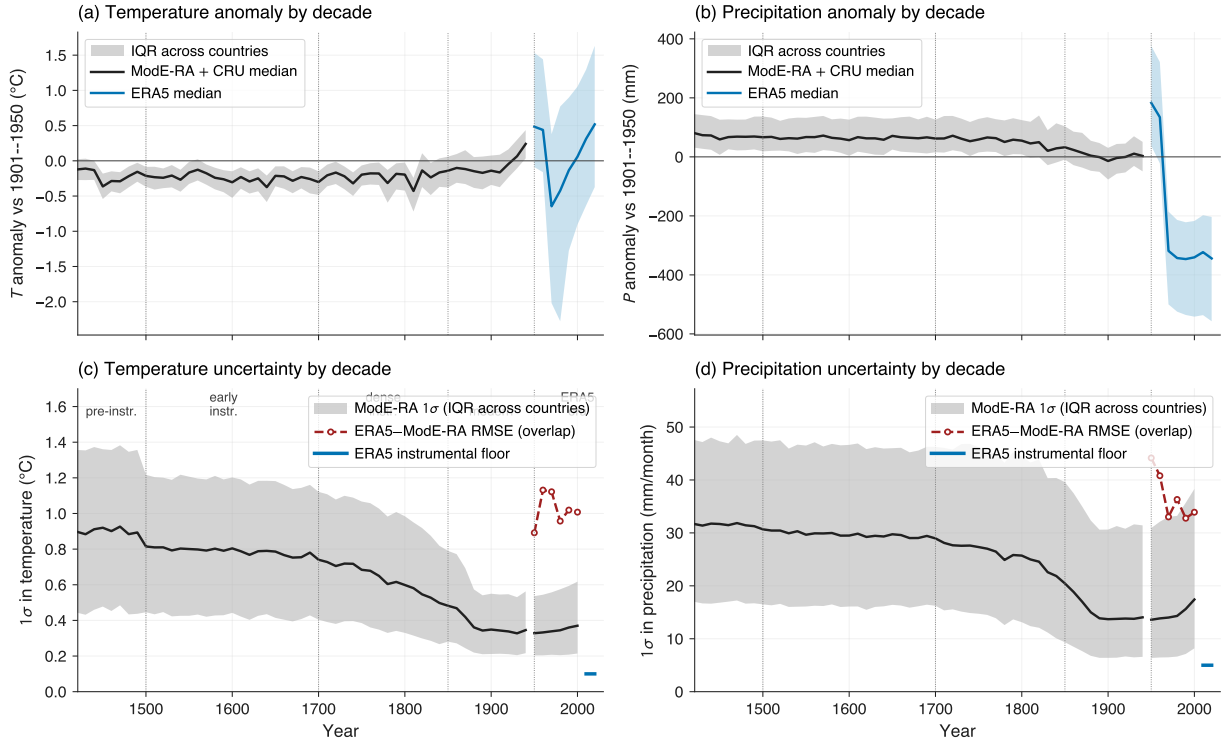


Figure 7: The combined ModE-RA + ERA5 climate panel, 1421–2025: *levels* (top row) and *uncertainty* (bottom row). (a) Temperature anomaly versus the 1901–1950 reference; (b) precipitation anomaly. (c–d) Country-month 1σ uncertainty by decade: grey band is ModE-RA’s published ensemble standard deviation; red dashed line is the ERA5–ModE-RA cross-product RMSE in the 1950–2008 overlap; blue stub at 2010–2020 is the literature-based ERA5 instrumental floor. The lower envelope is the working uncertainty propagated into the regressions.

H Deep-determinants robustness

A standard objection to cross-country long-run regressions is that the predictor proxies for a deep determinant of development—distance from a Neolithic origin centre, ancestry-adjusted genetic diversity, state antiquity, landlocked-ness, or terrain ruggedness—rather than the channel the paper claims. We add the standard battery: haversine distance to the nearest of seven Neolithic origin centres, migratory distance from Addis Ababa, a landlocked dummy, log land area, a centroid-based ruggedness proxy, mean pre-industrial temperature, and absolute latitude. Added sequentially, they leave the $\sigma_v^T \rightarrow \log \text{pop growth}$ coefficient remarkably stable.

Table 21: Long-shadow with deep-determinants battery: $\sigma_v^T \rightarrow \log \text{pop growth}$, $N = 199$ countries.

Specification	$\hat{\beta}_{\sigma_v^T}$	SE	p	R^2
Baseline	−2.48	0.24	$< 10^{-24}$	0.19
+ lat	−1.88	0.56	$< 10^{-3}$	0.20
+ lat + log area + landlocked	−2.08	0.55	$< 10^{-3}$	0.25
+ lat + Neolithic dist + Addis dist	−2.35	0.59	$< 10^{-4}$	0.34
+ all extended deep determinants	−2.09	0.53	$< 10^{-4}$	0.38

Robust standard errors. Extended battery: latitude, log area, landlocked, log distance to nearest Neolithic centre, log migratory distance from Addis Ababa, ruggedness proxy, and mean pre-industrial temperature.

Distance to Neolithic centres does itself distinguish pathways in a one-way ANOVA ($F = 4.96$, $p = 0.003$), but productive-months is the stronger Stage 1 predictor ($F = 6.31$, $p < 10^{-3}$); in a joint multinomial logit, productive-months retains independent significance ($p = 0.001$ in the crop-dominant equation) after controlling for Neolithic distance. Interacting Sigl-Toohey VSSI with standardised log distance to Neolithic centres in the volcanic regression returns an interaction coefficient of $-3.6 \cdot 10^{-6}$ with $p = 0.57$: the pathway-heterogeneous volcanic response is not driven by countries’ deep-historical exposure to agriculture.

I A calibrated three-equation quantitative model

We calibrate a three-equation pathway-specific dynamic model that takes the joint-VAR coefficients of Section 4.1 as structural inputs and forward-simulates the demographic, cropland-share, and urban-share trajectories under three counterfactuals. The model is parsimonious by design: we keep only the pathway-specific drift and the volcanic-forcing coefficient in each equation, calibrating the drift to match each pathway’s pre-industrial mean growth rate after subtracting the mean-volcanic contribution. The exercise is calibration, not deep structural estimation.

For country i in pathway τ and HYDE interval $[t, t + 1)$,

$$\Delta_{\text{ann}} \ln N_{it} = \alpha_{\tau}^P + \phi_{\tau}^P V_{it} + \varepsilon_{it}^P, \quad (7)$$

$$\Delta_{\text{ann}} \ln s_{it} = \alpha_{\tau}^s + \phi_{\tau}^s V_{it} + \varepsilon_{it}^s, \quad (8)$$

$$\Delta_{\text{ann}} \ln u_{it} = \alpha_{\tau}^u + \phi_{\tau}^u V_{it} + \varepsilon_{it}^u, \quad (9)$$

where V_{it} is the decade-summed Sigl-Toohey VSSI exposure. The coefficients ϕ_{τ}^* come directly from the joint-VAR pathway-stratified slopes (Table 27 and Section 4.1); the drifts α_{τ}^* are calibrated as $\alpha_{\tau}^* = \overline{g_{\tau, \text{pre1750}}} - \phi_{\tau}^* \overline{V_{\text{pre1750}}}$ so the simulation reproduces each pathway’s observed pre-industrial mean growth rate when V_{it} is at its pre-industrial mean.

Table 22: Calibrated parameters for the three-equation model, per pathway. α^* in annualised log-units; ϕ^* in annualised log-units per Tg of VSSI.

Pathway	α^P	ϕ^P	α^s	ϕ^s	α^u	ϕ^u
Crop-dominant late	$+8.5 \cdot 10^{-3}$	$-9.6 \cdot 10^{-5}$	$-3.0 \cdot 10^{-4}$	$+3.0 \cdot 10^{-5}$	$+2.2 \cdot 10^{-2}$	$-5.2 \cdot 10^{-4}$
Pastoral/mixed late	$+5.5 \cdot 10^{-3}$	$-5.5 \cdot 10^{-5}$	$-5.7 \cdot 10^{-4}$	$+1.3 \cdot 10^{-6}$	$+1.1 \cdot 10^{-2}$	$-6.0 \cdot 10^{-5}$
High-density intensive	$+4.2 \cdot 10^{-3}$	$-2.0 \cdot 10^{-5}$	$+2.0 \cdot 10^{-3}$	$-3.7 \cdot 10^{-5}$	$+8.7 \cdot 10^{-3}$	$+1.8 \cdot 10^{-5}$
Early extensifiers	$+1.7 \cdot 10^{-3}$	$-1.4 \cdot 10^{-5}$	$+1.3 \cdot 10^{-3}$	$-1.7 \cdot 10^{-5}$	$+1.2 \cdot 10^{-2}$	$-7.8 \cdot 10^{-5}$

The cropland-share coefficient ϕ^s is reproduced from the joint VAR; Section 4.5 documents that this coefficient is measurement-driven, so the cropland-share column of the counterfactual below is partial-equilibrium arithmetic of a HYDE-fitted coefficient, not a structural identification.

Two counterfactuals on the three-equation model. CF1 reassigns pastoral/mixed-late countries to the high-density-intensive coefficient vector and re-simulates. CF2 sets $V_{it} = 0$ in every cell.

Table 23: Counterfactual cumulative log-changes 1500–1890, median country per pathway. “ Δ vs baseline” rows report the counterfactual minus the baseline log-change.

Pathway	Specification	$\Delta \ln P$	$\Delta \ln s$	$\Delta \ln u$
Crop-dominant late	Baseline	+2.26	+0.23 (+26%)	+2.79
	CF2 no volcanism	+3.40	−0.12 (−12%)	+8.88
Pastoral/mixed late	Baseline	+1.54	−0.21 (−19%)	+3.69
	CF1 → intensive params	+1.96	−0.67 (−49%)	+3.69
	CF2 no volcanism	+2.20	−0.23 (−21%)	+4.40
High-density intensive	Baseline	+1.46	+0.36 (+43%)	+3.69
	CF2 no volcanism	+1.70	+0.80 (+123%)	+3.48
Early extensifiers	Baseline	+0.50	+0.33 (+39%)	+3.82
	CF2 no volcanism	+0.67	+0.53 (+70%)	+4.75

Cumulative log-changes from forward-simulating Equations (7)–(9) over each cell in the 1500–1890 panel; medians taken across countries within pathway. Percent labels next to $\Delta \ln s$ values convert log-units to level-change percentages, $\exp(\Delta) - 1$ (cropland share at 1890 relative to 1500). Population columns omit percent labels because cumulative log-units over four centuries produce uninformative level-change percentages.

CF2 is the substantive demographic counterfactual: crop-dominant late countries lose +2.12 log-units of cumulative population growth in the no-volcanism world (a factor of ~ 3 in level terms), pastoral/mixed late countries lose a similar amount in proportional terms, intensive and extensifier pathways less. Using the Allen Malthus regression of §4.3 ($\hat{\beta}_{\ln \text{pop}} = -0.285$), the per-capita welfare gain for crop-dominant late under CF2 is -0.32 log-units—the Malthusian welfare paradox: removing the volcanic check produces more people each substantially poorer.

The model is calibration, not deep structural estimation: lagged levels are treated as exogenous shifters and cross-equation feedback is set to zero (those coefficients are small in the joint VAR). The model does not reproduce the cross-country long-shadow regression ($\hat{\beta}^{\text{sim}} = +0.18$, $p = 0.56$): the cross-country long shadow operates through institutional and technological mediators that within-country dynamics do not capture. The cropland-share column should be read as partial-equilibrium arithmetic of a HYDE-fitted coefficient (Section 4.5), not as a structural identification.

J HYDE cropland measurement diagnostics and KK10 cross-validation details

This appendix collects the supporting tables and figures for the cropland-margin honest null of Section 4.5: the measurement-robustness diagnostics on the HYDE cropland-share specification (Table 24), the orthogonality test on the cropland-area outcome (Table 25), the full HYDE-vs-KK10 cross-validation table across windows (Table 26), and the country-level test of the substitution pattern between demographic and Boserupian margins (Figure 8).

Table 24: Cropland-margin response to VSSI: measurement-robustness diagnostics. The headline $+7.8 \cdot 10^{-4}$ cropland-share coefficient at 1800–1900 does not survive (i) substitution of the cropland-share outcome by its absolute area numerator and denominator; (ii) restriction to the substantively-named 21-country core; (iii) population weighting. Country FE, country-clustered SEs, joint-VAR controls throughout.

Specification	N	$\hat{\phi}$ per Tg	SE	p
<i>Panel A. Alternative cropland outcomes, 45 countries, 1800–1900:</i>				
$\Delta \ln s$ (cropland share, headline)	450	$+7.8 \cdot 10^{-4}$	$(2.6 \cdot 10^{-4})$	0.003
$\Delta \ln$ cropland area (km ²)	405	$-9.6 \cdot 10^{-5}$	$(5.4 \cdot 10^{-5})$	0.077
$\Delta \ln$ grazing area (km ²)	405	$-7.0 \cdot 10^{-3}$	$(2.3 \cdot 10^{-3})$	0.002
$\Delta \ln$ cropland per capita	405	$+9.0 \cdot 10^{-6}$	$(1.8 \cdot 10^{-5})$	0.613
<i>Panel B. 21-country Mediterranean / Eurasian-steppe / SE-Asia core, 1800–1900:</i>				
$\Delta \ln s$ (cropland share)	210	$+1.1 \cdot 10^{-5}$	$(9.6 \cdot 10^{-6})$	0.242
$\Delta \ln$ cropland area (km ²)	189	$+2.0 \cdot 10^{-6}$	$(5.5 \cdot 10^{-5})$	0.971
$\Delta \ln$ grazing area (km ²)	189	$-6.2 \cdot 10^{-5}$	$(7.0 \cdot 10^{-5})$	0.376
$\Delta \ln$ cropland per capita	189	$+4.8 \cdot 10^{-5}$	$(3.6 \cdot 10^{-5})$	0.183
<i>Panel C. Pop-weighted estimation, 45 countries, $\Delta \ln s$:</i>				
1700–1900 (post-century cells)	900	$+3.3 \cdot 10^{-5}$	$(1.9 \cdot 10^{-5})$	0.082
1800–1900 (modern instrumental)	450	$+6.1 \cdot 10^{-5}$	$(3.8 \cdot 10^{-5})$	0.103
<i>Panel D. Non-overlapping windows, 45 countries, $\Delta \ln s$:</i>				
1700–1750	225	$+3.3 \cdot 10^{-5}$	$(2.3 \cdot 10^{-5})$	0.153
1750–1800	225	0 (precisely)	$(1.5 \cdot 10^{-7})$	0.296
1800–1900	450	$+7.8 \cdot 10^{-4}$	$(2.6 \cdot 10^{-4})$	0.003
Cochran Q heterogeneity test		$Q = 10.83$ on 2 d.f.	$I^2 = 82\%$	$p = 0.004$

Panel A decomposes the cropland-share outcome of (6) into its numerator (log cropland area in km²) and denominator components, plus the per-capita measure. Panel B drops 24 small-island and tropical-Africa countries (Brunei, Côte d’Ivoire, Cameroon, Dominica, Equatorial Guinea, Fiji, Guadeloupe, Guinea-Bissau, Haiti, Jamaica, Lesotho, Lucia, Mauritius, Martinique, New Caledonia, Réunion, Saint-Kitts, Sierra Leone, Singapore, El Salvador, Tonga, Vanuatu, Vincent, Samoa) whose HYDE cropland and grazing trajectories include large 1840, 1850, and 1890 colonial-transition discontinuities. Panel C reports population-weighted least squares. Panel D splits the post-1700 panel into three disjoint blocks; the 1750–1800 cluster-robust SE is mechanically tight on a precisely-zero coefficient.

Table 25: Cropland-area response to volcanic forcing, with and without $\Delta_{\text{ann}} \ln P$ as control. The area response is statistically robust in baseline specifications but collapses to zero once the demographic margin is partialled out.

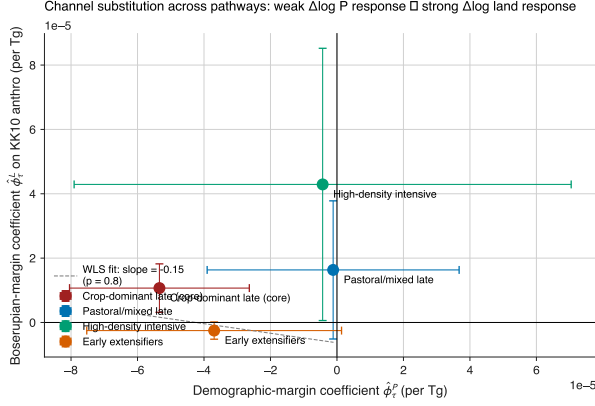
Sample	Window	Baseline $\hat{\phi}^{\text{area}}(p)$	$+\Delta \log P$ control (p)
Crop-dominant late, full 45-country	1500–1900	$-1.0 \cdot 10^{-4}$ ($< 10^{-6}$, ***)	$-7.8 \cdot 10^{-6}$ (0.24)
	Post-1700	$-2.6 \cdot 10^{-5}$ (0.14)	$+6.5 \cdot 10^{-6}$ (0.46)
	1800–1900	$-9.6 \cdot 10^{-5}$ (0.08, *)	$+4 \cdot 10^{-7}$ (0.99)
Crop-dominant late, 21-country core	1500–1900	$-3.5 \cdot 10^{-5}$ (0.005, ***)	$+7 \cdot 10^{-7}$ (0.91)
	Post-1700	$-6.1 \cdot 10^{-5}$ (0.001, ***)	$+8 \cdot 10^{-7}$ (0.96)
	1800–1900	$+2 \cdot 10^{-6}$ (0.97)	$+4.9 \cdot 10^{-5}$ (0.18)
Pastoral/mixed late, full 65-country	1500–1900	$-8.5 \cdot 10^{-5}$ ($< 10^{-3}$, ***)	$-5.2 \cdot 10^{-5}$ (0.18)
High-density intensive, full 16-country	1500–1900	$-3.0 \cdot 10^{-5}$ (0.05, *)	$-1.0 \cdot 10^{-5}$ (0.12)
Early extensifiers, full 27-country	Post-1700	$-4.3 \cdot 10^{-5}$ (0.027, **)	$-9 \cdot 10^{-7}$ (0.92)

Outcome: $\Delta_{\text{ann}} \ln(\text{cropland km}^2)$. Country FE, country-clustered SEs, joint-VAR controls. The “ $+\Delta \log P$ control” column adds the annualised log population growth to the right-hand side. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

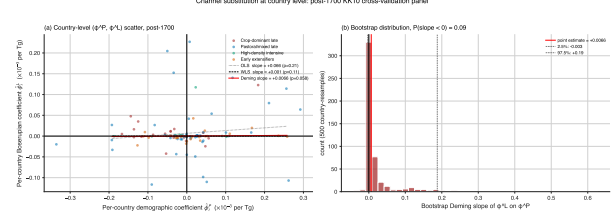
Table 26: HYDE vs. KK10 cross-validation of the area-orthogonality finding, 21-country crop-dominant-late core. The HYDE area response is negative and collapses under $\Delta_{\text{ann}} \ln P$ control. The KK10 area response is positive and survives the same control, identifying a small but statistically robust Boserupian extensive-margin response on the population-independent reconstruction.

Sample / window	Reconstruction	Baseline $\hat{\phi}(p)$	$+\Delta \log P$ control (p)	Survives
Post-1700 21-c core	HYDE crop+past area	$-5.96 \cdot 10^{-5}$ (0.0008, ***)	$+2.4 \cdot 10^{-6}$ (0.85)	no
Post-1700 21-c core	KK10 anthro area	$+1.07 \cdot 10^{-5}$ (0.005, ***)	$+1.08 \cdot 10^{-5}$ (0.005, ***)	yes
Post-1750 21-c core	HYDE crop+past area	$-1.49 \cdot 10^{-5}$ (0.42)	$+3.4 \cdot 10^{-6}$ (0.82)	—
Post-1750 21-c core	KK10 anthro area	$+1.52 \cdot 10^{-5}$ (0.011, **)	$+1.53 \cdot 10^{-5}$ (0.007, ***)	yes
Full 1500–1850 21-c core	HYDE crop+past area	$-3.02 \cdot 10^{-5}$ (0.011, **)	$+5.5 \cdot 10^{-6}$ (0.42)	no
Full 1500–1850 21-c core	KK10 anthro area	$+1.1 \cdot 10^{-6}$ (0.80)	$+2.8 \cdot 10^{-6}$ (0.55)	—
1800–1850 21-c core	HYDE crop+past area	$-1.26 \cdot 10^{-5}$ (0.81)	$+3.3 \cdot 10^{-5}$ (0.25)	—
1800–1850 21-c core	KK10 anthro area	$-1.11 \cdot 10^{-5}$ (0.40)	$-1.06 \cdot 10^{-5}$ (0.40)	—

Outcomes are annualised log-change of combined anthropogenic land (cropland + grazing) in km^2 . KK10 ends at 1850, so the comparison windows are truncated by 4 decades relative to the HYDE post-1700 panel. “Survives” = baseline coefficient is significant at $p < 0.05$ and remains so after adding $\Delta_{\text{ann}} \ln P$ as a control.



(a) Pathway-level scatter ($N = 4$). WLS slope -0.15 ($p = 0.80$): the qualitative substitution pattern is visible, but the slope is not statistically distinguishable from zero with four points.



(b) Country-level scatter ($N = 116$) and Deming-regression bootstrap. Deming slope $+0.007$ ($p = 0.06$), bootstrap 95% CI $[-0.003, +0.19]$, 91% positive. The country-level test does not identify a substitution slope.

Figure 8: Channel substitution across pathways (panel a) and across countries (panel b). The pathway-level scatter is illustrative; the country-level test does not confirm a substitution slope.

For each of 116 countries with both coefficients estimable on a post-1700 reduced-form regression (lagged log population, interval-mean temperature, within-interval temperature volatility, VSSI, with country intercept), we extract $(\hat{\phi}_i^P, \hat{\phi}_i^L)$ and run an errors-in-variables (Deming) regression across countries, weighted by $1/\text{SE}^2$ on both axes. The Deming slope is $+0.007$ ($\text{SE} = 0.004$, $p = 0.06$); a 500-replication country-resampling bootstrap returns a 95% confidence interval $[-0.003, +0.19]$ with 91% of bootstrap slopes positive. The country-level test does not identify a negative substitution slope; if anything, the country-level relationship is weakly positive (countries with stronger demographic contractions also show slightly more land-use contraction)—consistent with joint climate vulnerability rather than channel substitution. At country level, the noise of decadal panel time series at 15–18 observations per country dominates whatever cross-country variation in margin choice exists.

K Robustness of the cross-pathway asymmetry

This appendix documents the three robustness exercises that the main-text Section 6 summarises. They target the joint-VAR cross-pathway asymmetry of Section 4.1 specifically, which depends on a pathway typology estimated from HYDE-trajectory outcomes (peak agricultural expansion year, 1750-era density, 1750-era crop share, urbanisation timing).

L.1 Climate-only pathway clustering

The headline pathway typology of Section 2.3 is built on HYDE-trajectory features that are downstream outcomes of the same processes the joint VAR is trying to explain. We re-cluster the 196 countries using exclusively pre-industrial climate primitives—productive months, intra-annual T and P range, mean T , mean P , and inter-annual T volatility, all computed over 1421–1750 from the Mode-RA + CRU panel and orthogonal to every HYDE outcome—and re-estimate the joint VAR on the new cluster labels.

K=5 KMeans on the standardised climate-primitive features achieves a silhouette of 0.340 and produces five clusters ordered by mean productive months: *Low-storage (driest)* (mean prod. months 2.5, $\sigma_s = 15.4^\circ\text{C}$, $N = 32$), *Moderate-storage low* (4.1, 28.1°C , $N = 22$), *Mid-storage* (7.1, 20.7°C , $N = 33$), *Moderate-storage high* (9.1, 7.0°C , $N = 49$), and *High-storage (wettest)* (11.1, 3.9°C , $N = 60$). The cross-tab against the HYDE-derived clusters is essentially block-diagonal-free: HYDE-cluster “pastoral/mixed late” (the largest, $N = 65$) spreads across all five climate clusters, and similarly for HYDE-cluster “crop-dominant late”. The two typologies measure different things.

Table 27: Joint VAR pathway-stratified VSSI slopes under climate-only clustering, 1500–1900. Country FE, country-clustered SEs. Stars: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Climate cluster (storage demand)	N	countries	$\hat{\beta}^{\text{Pop}} (p)$	$\hat{\beta}^{\text{Crop}} (p)$	$\hat{\beta}^{\text{U}}$
Low-storage (driest)	483	22	$-2.6 \cdot 10^{-5} (0.048)**$	$-2.9 \cdot 10^{-5} (0.330)$	$-5.1 \cdot 10^{-5} (0.000)$
Moderate-storage low	308	14	$-3.2 \cdot 10^{-5} (0.045)**$	$+6.9 \cdot 10^{-6} (0.250)$	$+5.6 \cdot 10^{-5} (0.000)$
Mid-storage	704	32	$-3.0 \cdot 10^{-5} (0.007)***$	$-3.9 \cdot 10^{-5} (< 10^{-5})***$	$-4.7 \cdot 10^{-5} (0.000)$
Moderate-storage high	814	37	$-7.4 \cdot 10^{-5} (< 10^{-3})***$	$-2.9 \cdot 10^{-5} (0.008)***$	$-4.8 \cdot 10^{-5} (0.000)$
High-storage (wettest)	1,076	49	$-8.1 \cdot 10^{-5} (< 10^{-6})***$	$+5.4 \cdot 10^{-5} (0.091)*$	$-5.2 \cdot 10^{-4} (0.000)$

Pathways labeled by mean pre-industrial productive months and intra-annual temperature range. The neat one-pathway-per-margin pattern of the HYDE-clustered specification (Section 4.1) does not survive: all five climate-only clusters show statistically significant negative VSSI coefficients on $\Delta \log P$, and the cropland and urban responses are not concentrated in single climate-defined clusters.

L.2 Wild-cluster bootstrap and label permutation

With twelve pathway-by-equation slopes in the headline joint VAR (four informative pathways times three equations), the asymptotic single-coefficient significance tests are vulnerable to multiple-testing inflation. We re-test the twelve slopes two ways.

First, a Webb (2014) six-point wild-cluster bootstrap (1,999 replications, restricted residuals re-signed at the country level) gives bootstrap p-values that track the cluster-robust asymptotic p-values closely (the two columns of Table 28 agree to within rounding). The cluster SEs are not catastrophically off; the asymptotic inference is well-calibrated. Under Bonferroni correction at the 5% level ($0.05/12 = 0.0042$), however, only two of the twelve slopes survive: the crop-dominant late and pastoral/mixed late demographic-margin coefficients (Table 28). The high-density intensive cropland coefficient ($p_{\text{boot}} = 0.017$) and the early-extensifier urban coefficient ($p_{\text{boot}} = 0.035$) fall short of Bonferroni significance, as do all remaining off-margin coefficients. A Benjamini-Hochberg FDR procedure at the 5% level returns the same two demographic-margin coefficients.

Table 28: Wild-cluster bootstrap p-values for the joint VAR pathway slopes, 1,999 Webb six-point replications. Bonferroni threshold: $p < 0.0042$. Stars: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Pathway	Equation	$\hat{\beta}$	p_{asympt}	p_{boot}	Bonferroni
Crop-dominant late	$\Delta \log P$	$-9.65 \cdot 10^{-5}$	$< 10^{-6}$ ***	0.000 ***	✓
Crop-dominant late	$\Delta \log s$	$+2.96 \cdot 10^{-5}$	0.258	0.301	
Crop-dominant late	$\Delta \log u$	$-5.16 \cdot 10^{-4}$	0.126	0.045 **	
Pastoral/mixed late	$\Delta \log P$	$-5.55 \cdot 10^{-5}$	$< 10^{-5}$ ***	0.000 ***	✓
Pastoral/mixed late	$\Delta \log s$	$+0.13 \cdot 10^{-5}$	0.913	0.931	
Pastoral/mixed late	$\Delta \log u$	$-6.02 \cdot 10^{-5}$	0.086 *	0.280	
High-density intensive	$\Delta \log P$	$-2.02 \cdot 10^{-5}$	0.215	0.233	
High-density intensive	$\Delta \log s$	$-3.73 \cdot 10^{-5}$	0.021 **	0.017 **	
High-density intensive	$\Delta \log u$	$+1.84 \cdot 10^{-5}$	0.751	0.823	
Early extensifiers	$\Delta \log P$	$-1.43 \cdot 10^{-5}$	0.121	0.146	
Early extensifiers	$\Delta \log s$	$-1.71 \cdot 10^{-5}$	0.186	0.210	
Early extensifiers	$\Delta \log u$	$-7.82 \cdot 10^{-5}$	0.040 **	0.035 **	

Second, a permutation test that re-shuffles the pathway labels across countries 300 times—preserving panel structure and within-country dynamics but breaking the link between cluster identity and country—asks how often a random labeling produces at least the observed number of significant coefficients per equation. For the demographic equation the answer is essentially always (median 4 significant slopes under permutation vs. 2 observed); the labeling actually reduces the count of significant demographic slopes relative to random. For the cropland equation, 48% of permutations produce at least one significant cropland slope. For the urban equation, 73% of permutations produce at least one significant urban slope. The observed cross-pathway pattern of one significant cropland and one significant urban response across four pathways is not distinguishable from what random labels produce.

L.3 Post-1700 sub-sample

HYDE 3.5 uses 100-year intervals before 1700 and decadal intervals afterwards; the 1500 and 1600 cells in the joint VAR panel are century-aggregated back-projections from a sparser historical source base than the post-1700 decadal cells. HYDE’s pre-1850 demographic back-projection also uses cropland and pasture extent as input variables, so the joint cropland-and-population dynamics in the pre-1700 portion of the panel are partly mechanical. We restrict to year ≥ 1700 (drops 305 century cells, $\sim 9\%$ of the panel) and re-run.

The post-1700 restriction differentially weakens the pathway coefficients. Table 29 reports the slopes across four sub-samples: full 1500–1900, post-1700, post-1750, and post-1800. The crop-dominant late demographic margin survives every cut ($p \leq 0.018$). The pastoral/mixed demographic margin collapses to $p = 0.95$ post-1700: that significance came entirely from the century-resolution cells. The high-density intensive cropland margin weakens to $p = 0.076$ post-1700 and to $p = 0.26$ post-1750. The early-extensifier urban margin weakens to $p = 0.28$ post-1700.

Table 29: Joint VAR pathway slopes across temporal sub-samples. Coefficients in units of 10^{-5} per Tg VSSI. Stars: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Pathway	Sub-sample	$\hat{\beta}^{\text{Pop}} (p)$	$\hat{\beta}^{\text{Crop}} (p)$	$\hat{\beta}^{\text{Urb}} (p)$
Crop-dominant late	1500–1900 (full)	−9.65 ***	+2.96	−51.6
	1700–1900	−4.10 **	+36.3 **	+9.95
	1750–1900	−5.66 ***	+39.1 **	+0.19
	1800–1900	−11.9 **	+78.4 ***	−0.13
Pastoral/mixed late	1500–1900 (full)	−5.55 ***	+0.13	−6.02 *
	1700–1900	−0.11	−4.79	−0.61
	1750–1900	−0.48	−5.99	+0.79
	1800–1900	−2.22	−10.0	−2.45
High-density intensive	1500–1900 (full)	−2.02	−3.73 **	+1.84
	1700–1900	−0.43	−4.16 *	−8.44
	1750–1900	−0.23	−3.32	−9.18
	1800–1900	−5.94 *	−2.26	−4.81
Early extensifiers	1500–1900 (full)	−1.43	−1.71	−7.82 **
	1700–1900	−3.69 *	+10.7	−3.13
	1750–1900	−2.93	+15.7 *	−2.82 *
	1800–1900	−6.11	+20.7	−3.38

The crop-dominant late demographic margin is the only coefficient that retains significance across all four sub-samples. The other three margins of the original three-margin asymmetry either weaken substantially (high-density intensive cropland, early extensifier urban) or vanish (pastoral/mixed demographic) when the century-resolution HYDE cells are dropped.

L.4 Reading the robustness exercises

The three exercises target related but distinct issues with the joint-VAR cross-pathway asymmetry: pathway endogeneity (L.1), multiple-testing inflation (L.2), and HYDE temporal-imputation artifacts (L.3). Each exercise weakens a different component of the original headline. Taking the three together, what is statistically defensible is a clean demographic-margin response to volcanic forcing that is concentrated in crop-dominant late countries and is robust across pathway typology, multiple-testing correction, and HYDE sub-sample. What is not statistically defensible at conventional levels is the further claim that intensive and early-extensifier pathways adjust on non-demographic margins: those slopes are individually nominally significant in the full HYDE-clustered specification but do not survive any one of the three robustness exercises.

A noteworthy and previously unreported result emerges from the post-1700 restriction: the crop-dominant late pathway shows a sharply *positive* cropland-share response to volcanic forcing once the century-resolution cells are dropped ($\hat{\beta} = +3.6 \cdot 10^{-4}$ per Tg post-1700, $p = 0.031$; $+7.8 \cdot 10^{-4}$, $p = 0.003$ in 1800–1900). After volcanic cooling, crop-dominant late economies expanded their cropland share—a Boserupian intensification response in the decade following a cold shock. This finding is robust to the temporal sub-sample but is masked in the full 1500–1900 panel by the offsetting pre-1700 century cells. We note it here as a candidate result for follow-up work.

The honest reading of the joint VAR is therefore: it identifies a clean demographic effect within a multi-margin framework, surfaces an unexpected Boserupian cropland response in the modern-era crop-dominant subsample, and motivates future work with finer temporal demographic resolution that can resolve the within-decade dynamics of the cropland and urban margins.

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